

Robust Nash Preference Optimization

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Abstract

Aligning language models increasingly relies on multiple, conflicting preference signals: helpfulness, harmlessness, conciseness, and domain-specific judges routinely disagree. Standard practice scalarizes these signals into a fixed average, which can hide a severe failure on a minority objective, while assigning each objective its own player yields a general-sum game for which last-iterate convergence guarantees are unavailable. We propose Robust Nash Preference Optimization (RONPO), which keeps heterogeneous alignment inside a two-player game: a single adversary jointly selects a preference oracle and an opponent response, and the policy must remain competitive against the resulting worst-case pair. The induced objective is a KL-regularized bilinear saddle whose inner problem is a distributionally robust soft minimum over objective–response pairs, so the policy maximizes a smooth conservative surrogate of its worst-case win probability, one that brackets the hard worst case to within an explicit temperature-dependent gap. We prove that the game admits a unique robust Nash policy and that exact mirror-descent dynamics converge directly to it at an $O(1/T)$ rate. The exact update reduces to a partition-free log-ratio regression against a sampled adversarial pair, giving a practical loss for language models. On synthetic preference games, RONPO nearly closes the gap to the robust linear-programming optimum where averaging-based methods collapse on a hidden objective, and its dynamics track the predicted convergence rate. At model scale, RONPO raises worst-objective reward under multiple judges and preserves verifiable instruction following. The model-scale evidence uses scalar reward models through a Bradley–Terry plug-in oracle, so it validates heterogeneous-reward robustness; genuinely non-transitive preferences are exercised only in the synthetic games.

Introduction

Reinforcement learning from human feedback (RLHF) is the standard recipe for aligning Large Language Models (LLMs): pairwise human preferences are compressed into a scalar reward model, and the policy is optimized against that reward under a KL constraint (Christiano et al. 2017; Ouyang et al. 2022). Direct alignment methods such as DPO

shortcut the reward-modeling step but retain the same scalar, Bradley–Terry structure (Rafailov et al. 2023; Bradley and Terry 1952). This compression is a modeling assumption, not a fact about human judgment: a Bradley–Terry model induces a transitive order over responses, whereas aggregate human preferences can be cyclic or context-dependent (Tversky 1969; May 1954).

When preferences need not be transitive, there may exist no scalar utility whose maximizer is “the preferred response,” so reward maximization is no longer the right solution concept. Nash learning from human feedback (NLHF) instead works with a general preference oracle and seeks a policy that no opponent policy can beat in expectation, a von Neumann winner of the induced two-player constant-sum game (Munos et al. 2024; Dudík et al. 2015; Azar et al. 2024). The game-theoretic view is what supplies robustness: the learned policy is evaluated against an explicit opponent that actively searches for responses on which the policy loses, rather than against a fixed scalar score. Self-play methods such as SPPO and INPO make this practical for language models via no-regret mirror descent, and INPO in particular provides *last-iterate* convergence to the regularized Nash policy (Wu et al. 2025; Zhang et al. 2025).

Realistic alignment, however, rarely involves a single preference source. A model is judged simultaneously by helpfulness, harmlessness, factuality, conciseness, style, and domain-policy critics (Bai et al. 2022; Wang et al. 2024c,b), and these judges conflict (Dai et al. 2024; Ramé et al. 2023). Existing methods absorb this conflict at the level of objectives alone: single-oracle methods such as SPPO and INPO may collapse the judges into one signal, typically a fixed weighted average (Wu et al. 2025; Zhang et al. 2025; Wang et al. 2024b), while robust variants such as MaxMin-RLHF, group-robust preference optimization, and RMOD harden only the choice of objective weights (Chakraborty et al. 2024; Ramesh et al. 2024; Son et al. 2026). Assigning each oracle its own player, as in MNPO (Wu et al. 2026), avoids aggregation altogether, but the game then becomes general-sum and the monotone structure behind last-iterate convergence is lost. What is needed is a method that is robust jointly over objectives and opponent responses while remaining a two-player game with a last-iterate convergence guarantee.

RONPO keeps the multi-objective concept but changes where it lives. There is one policy player and one adver-

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sary, and the adversary’s action is a *pair* (k, a) : which preference oracle judges the policy, and which opponent response the policy must beat. Confining conflicting preferences to a single adversary’s action set keeps the problem a two-player KL-regularized bilinear saddle, so the strongly monotone structure (and with it existence, uniqueness, and last-iterate convergence) survives arbitrary conflict among the K oracles. The inner minimization is the penalty form of a Kullback–Leibler distributionally robust program over pairs: RONPO maximizes an entropic soft minimum of the objective–response win probabilities, a smooth conservative surrogate that brackets the hard worst case to within an explicit κ -dependent gap (Prop. 1), with the single temperature κ interpolating between hard worst case and plain averaging under the prior.

Our contributions are as follows.

- We cast multiple preference alignment as an objective–adversarial game whose adversary selects objective–response pairs.
- The game has a unique robust Nash policy (Thm. 1); exact mirror-descent dynamics converge to it at an $O(1/T)$ last-iterate rate (Thm. 2). To our knowledge, these are the first last-iterate guarantees for a general-preference formulation whose single adversary ranges jointly over heterogeneous preference oracles and opponent responses; concurrent multi-objective work targets scalar rewards, decoding-time robustness, or offline constrained formulations without a game-theoretic rate (see Related Work).
- The exact update reduces to a partition-free log-ratio regression against a sampled adversarial pair (Prop. 3), yielding a loss with no partition function.
- Synthetic preference games isolate the hidden-objective failure of averaging, where RONPO nearly closes the gap to the robust LP optimum at essentially no average cost, and numerically validate the convergence theory. At model scale, RONPO raises worst-objective reward under the training-aligned judges and preserves instruction following.

We are explicit about scope. Algebraically, RONPO is the standard KL-regularized two-player game on an adversary action set enlarged to objective–response pairs, and the equilibrium and rate analysis instantiates standard strongly monotone variational-inequality arguments (Facchinei and Pang 2003; Nemirovski and Yudin 1983). The contribution is what the joint index adds on top of this lifting: a formulation that keeps arbitrarily conflicting objectives inside a monotone two-player game, a finite-temperature approximation guarantee with an explicit gap (Prop. 1), a strict separation from weight-only adversaries (Prop. 2), and a stratified estimator with a variance-reduction guarantee behind the practical loss (Prop. 5).

Related Work

RLHF and direct alignment. Reward-based RLHF optimizes a KL-regularized policy against a learned scalar reward (Christiano et al. 2017; Ouyang et al. 2022). DPO removes the explicit reward model through a closed-form

reparameterization (Rafailov et al. 2023); IPO replaces its logistic link with a squared loss on preference probabilities and, by design, optimizes a general pairwise preference without any Bradley–Terry assumption (Azar et al. 2024); and SimPO drops the reference model (Meng, Xia, and Chen 2024). What separates this family from Nash alignment is therefore not transitivity but the opponent: preferences are maximized against offline data or a fixed reference distribution rather than against a worst-case response. Under heterogeneous judges these methods also consume a single or averaged signal, the regime our synthetic study shows can mask minority-objective collapse.

General preferences and Nash alignment. Dueling and contextual dueling bandits introduced learning from pairwise comparisons and the von Neumann winner solution concept (Yue et al. 2012; Dudík et al. 2015). NLHF brought this view to LLM alignment (Munos et al. 2024), with self-play and no-regret instantiations including SPPO, INPO, DNO, SPO, and online IPO variants (Wu et al. 2025; Zhang et al. 2025; Rosset et al. 2024; Swamy et al. 2024; Calandriello et al. 2024), and theoretical treatments of the KL-regularized preference game (Ye et al. 2024). EGPO strengthens this toolkit with extragradient updates that give last-iterate *linear* convergence for the single-oracle regularized game (Zhou, Fazel, and Du 2025); we discuss after Theorem 2 why RONPO nevertheless deploys plain mirror descent. MNPO extends NLHF to population opponents and, in its heterogeneous variant, to per-player oracles (Wu et al. 2026). RONPO reuses the mirror-descent foundation but selects heterogeneous objectives adversarially rather than averaging them; unlike the per-player split, it retains the two-player monotone structure (Lemma 1).

Robust and multi-objective alignment. MaxMin-RLHF optimizes an egalitarian max-min over scalar reward models (Chakraborty et al. 2024), group-robust preference optimization minimizes a worst-group DPO-style loss (Ramesh et al. 2024), MODPO folds multiple objectives into a single direct loss (Zhou et al. 2024), and RMOD computes maxmin weights over reward-model values at decoding time (Son et al. 2026). Two concurrent works are closest in motivation: MOPO casts multi-objective alignment as offline constrained optimization directly on pairwise preference data, maximizing a primary objective under lower bounds on secondary ones (Agnihotri et al. 2026), and Latham and Moharrami analyze offline constrained RLHF in which each objective is estimated from its own preference oracle, with finite-sample guarantees (Latham and Moharrami 2026). Both are constrained scalarizations solved against fixed offline data: neither models an adversarial opponent response, and their guarantees are statistical rather than game-theoretic, whereas RONPO trains against a jointly adversarial judge–response pair with a last-iterate equilibrium rate. A separate line characterizes the Pareto front itself, by weight interpolation (Ramé et al. 2023; Wang et al. 2024a) or with a single preference-conditioned model (Zhong et al. 2024). Since a deployed model ultimately commits to one policy, RONPO is one conservative selection criterion among several; we do not claim the resulting policy is Pareto-efficient for the vector

of mean rewards, and a response-level adversary with policy KL regularization need not land on that front. RONPO differs from the robust family on two axes: it works with general preference *oracles* rather than scalar rewards, retaining the intransitivity that motivates NLHF, and its adversary ranges over opponent responses a in addition to objective indices k , so it can expose failures that depend jointly on the judge and the opponent response (Prop. 2). For a fixed policy, the inner problem is exactly a KL-constrained DRO program over objective–response distributions (Hu and Hong 2013).

Preliminaries

Preference-Based Alignment as a Contextual Bandit

We work in the preference-based contextual bandit setting that underlies RLHF (Yue et al. 2012; Dudík et al. 2015). A context (prompt) $x \in \mathcal{X}$ is drawn from a distribution ρ ; the learner selects a response (action) y from a finite response set $\mathcal{Y}$; and feedback is *relative*: for a pair of responses (y, y') , a preference oracle returns a binary label

$$z \sim \text{Ber}(P(y \succ y' \mid x)), \quad z \in \{0, 1\}, \quad (1)$$

where $z = 1$ means y is preferred to y' . The oracle $P : \mathcal{Y} \times \mathcal{Y} \times \mathcal{X} \rightarrow [0, 1]$ is skew-symmetric, $P(y \succ y' \mid x) + P(y' \succ y \mid x) = 1$ and $P(y \succ y \mid x) = 1/2$, but need not be induced by any scalar utility: cyclic preferences are allowed. A policy maps each context to a distribution $\pi(\cdot \mid x) \in \Delta(\mathcal{Y})$, and for policies π, π' we write $P(\pi \succ \pi' \mid x) = \mathbb{E}_{y \sim \pi(\cdot \mid x), y' \sim \pi'(\cdot \mid x)}[P(y \succ y' \mid x)]$.

All objectives in this paper are expectations over $x \sim \rho$ of per-context terms, and the exact dynamics we analyze decompose across contexts. Following standard practice in NLHF analysis (Munos et al. 2024; Wu et al. 2025; Zhang et al. 2025), we therefore state definitions and results for a fixed context and drop x from the notation; every statement lifts to the contextual case by conditioning on x and averaging over $x \sim \rho$.

Nash Learning from General Preferences

With the context fixed, a policy is a probability vector $\pi \in \Delta(\mathcal{Y})$. The unregularized NLHF objective is $\max_{\pi} \min_{\pi'} P(\pi \succ \pi')$, whose solution is a von Neumann winner: a policy that no opponent policy beats in expectation (Munos et al. 2024; Dudík et al. 2015). KL regularization toward a fixed reference policy $\mu \in \text{int } \Delta(\mathcal{Y})$ (the role played by π_{ref} in direct alignment) yields the smoother saddle objective

$$J(\pi, \pi') = P(\pi \succ \pi') - \tau \text{KL}(\pi \parallel \mu) + \tau \text{KL}(\pi' \parallel \mu), \quad (2)$$

with the first player maximizing and the second minimizing, and $\tau > 0$ the regularization temperature. Because $P(\pi \succ \pi')$ is bilinear in (π, π') , the objective J is concave in π and convex in π' : (2) is a two-player zero-sum game in saddle-point form, and the regularized Nash policy is the π -component of its saddle point. What remains is a computational question, namely how to reach that saddle point, and the standard answer in NLHF is to let both players

run no-regret online algorithms against each other. We review this machinery next, since every method we compare, including ours, is an instance of it.

No-Regret Learning via Mirror Descent

Solving a saddle-point game online amounts to two coupled online convex problems: once one player’s iterate is fixed, the other faces a convex loss. If both players run no-regret algorithms, the duality gap of average play vanishes at rate $(R_T^\pi + R_T^{\pi'})/T$ (Freund and Schapire 1999; Cesa-Bianchi and Lugosi 2006). The standard no-regret rule on the simplex is entropic online mirror descent (OMD), the proximal update (39) whose closed form is the multiplicative-weights step (40) with regret $O(\sqrt{T} \log |\mathcal{Y}|)$; Appendix C states the update and collects the regret-to-gap reduction and the regret proof. Average-iterate convergence follows from regret alone; the last-iterate results below additionally exploit strong monotonicity of the regularized operator.

Existing Methods Through the OMD Lens

The self-play baselines instantiate template (40) and differ only in the opponent. **SPPO** (Wu et al. 2025) uses the pointwise self-play update $\pi_{t+1}(y) \propto \pi_t(y) \exp\{\eta(P(y \succ \pi_t) - \frac{1}{2})\}$ with $P(y \succ \pi_t) = \mathbb{E}_{a \sim \pi_t}[P(y \succ a)]$; under multiple preferences we instantiate it on an averaged oracle $\bar{P} = \sum_{k=1}^K w_k P_k$, where $P_1, \dots, P_K$ are per-objective preference oracles (formally introduced in the next section) and $w \in \Delta(\{1, \dots, K\})$ is a fixed weight vector (the SPPO-avg baseline of our experiments, with uniform weights). **INPO** (Zhang et al. 2025) applies OMD to the regularized game (2),

$$\pi_{t+1}(y) \propto \exp\{\eta P(y \succ \pi_t)\} \mu(y)^{\eta\tau} \pi_t(y)^{1-\eta\tau}, \quad (3)$$

and turns it into a partition-free loss by regressing pairwise log-ratios $\Delta_\pi(y, y') := \log \pi(y) - \log \pi(y')$, a device we reuse. **MNPO** (Wu et al. 2026) replaces the single opponent with a population via a geometric opponent mixture. Its heterogeneous extension assigns oracle P_i to player i ; the game is then neither symmetric nor constant-sum, the bilinear cross terms no longer cancel pairwise, and the operator need not be monotone. We do not claim such dynamics diverge; population play can perform well empirically (Wu et al. 2026; Ramé et al. 2023). Our claim is only that the monotone variational-inequality route to last-iterate convergence is unavailable. Table 1 places these methods and the robust family on common axes; RONPO recovers the monotone structure exactly (Lemma 1) by keeping a single adversary.

Robust Nash Preference Optimization

The Objective-Adversarial Game

Assume K preference oracles $P_1, \dots, P_K$ indexed by $\mathcal{K} = \{1, \dots, K\}$, and let $\mathcal{A} \subseteq \mathcal{Y}$ be a finite adversarial response pool. In LLM applications, $\mathcal{A}(x)$ is constructed from reference responses, current-policy responses, or historical checkpoints. The policy chooses $\pi \in \Delta(\mathcal{Y})$; the adversary chooses a distribution $s \in \Delta(\mathcal{K} \times \mathcal{A})$ over objective–response pairs (k, a) . Because $\mathcal{A}$ is finite, RONPO’s robustness statement is

Method	Oracle	Robustness axis	Train / decode	Opponent object	Guarantee (tabular)
DPO / IPO / SimPO	pairwise data	—	train	offline pair data	—
SPPO	general	—	train	current policy	average-iterate
INPO	general	—	train	current policy + reference	last-iterate $O(1/T)$
EGPO	general	—	train	current policy + reference	last-iterate, linear
MaxMin-RLHF	scalar rewards	objective weights	train	—	—
Group-robust PO	scalar (DPO loss)	group weights	train	offline pair data	—
RMOD	scalar rewards	objective weights	decode	—	—
MOPO	pairwise data	primary + constraints	train	offline pair data	—
MNPO (het.)	general, per player	per-player objectives	train	opponent population	— [†]
RONPO (ours)	general	joint objective–response pairs	train	adversarial pair	last-iterate $O(1/T)$

Table 1: Modeling comparison on common axes: whether the training signal is a scalar reward or a general pairwise oracle, which uncertainty the method is robust to, whether robustness acts at training or decoding time, the opponent object, and the strongest convergence guarantee we are aware of. Every listed guarantee, ours included, concerns idealized tabular dynamics and does not certify the deployed neural algorithm. [†]The heterogeneous (per-player oracle) MNPO variant is general-sum and falls outside the monotone class, so the homogeneous last-iterate argument does not transfer.

relative to this pool and to the K oracles: the learned policy is trained to be unexploitable by pairs in $\mathcal{K} \times \mathcal{A}$, not by every response in $\mathcal{Y}$.

Define the win functional

$$W(\pi, s) = \mathbb{E}_{y \sim \pi, (k,a) \sim s} [P_k(y \succ a)]. \quad (4)$$

Let $\mu \in \text{int } \Delta(\mathcal{Y})$ be the reference policy and $s_0 \in \text{int } \Delta(\mathcal{K} \times \mathcal{A})$ an adversary prior. RONPO solves

$$\begin{aligned} \max_{\pi \in \Delta(\mathcal{Y})} \min_{s \in \Delta(\mathcal{K} \times \mathcal{A})} V(\pi, s) &:= W(\pi, s) - \tau \text{KL}(\pi \| \mu) \\ &\quad + \kappa \text{KL}(s \| s_0), \end{aligned} \quad (5)$$

with temperatures $\tau, \kappa > 0$. Writing $A \in \mathbb{R}^{|\mathcal{Y}| \times K|\mathcal{A}|}$ for the matrix $A_{y,(k,a)} = P_k(y \succ a)$, we have $W(\pi, s) = \pi^\top A s$, and the saddle operator is

$$\begin{aligned} F(\pi, s) &= (-\nabla_\pi V(\pi, s), \nabla_s V(\pi, s)) \\ &= (-A s + \tau(\log(\pi/\mu) + \mathbf{1}), \\ &\quad A^\top \pi + \kappa(\log(s/s_0) + \mathbf{1})), \end{aligned} \quad (6)$$

with logarithms and divisions componentwise. If $K = 1$, $\mathcal{A} = \mathcal{Y}$, $s_0 = \mu$, and $\kappa = \tau$, then (5) reduces to the KL-regularized NLHF game (2), so RONPO strictly generalizes single-oracle NLHF.

Algebraically, (5) is nothing more exotic than the usual regularized bilinear game with the adversary’s action set enlarged from $\mathcal{Y}$ to $\mathcal{K} \times \mathcal{A}$: a reader who prefers to view (k, a) as one column index of A may do so, and the monotonicity, existence, and rate arguments below then run along standard lines. What the joint index contributes is semantic and statistical rather than algebraic: the inner problem acquires a distributionally robust reading over judge–response pairs with an explicit finite- κ approximation guarantee (Prop. 1), pair-level robustness is provably stronger than objective-weight robustness under a fixed opponent (Prop. 2), and the product structure of $\mathcal{K} \times \mathcal{A}$ admits a stratified estimator that gives every objective a training signal per update at cost K instead of $K|\mathcal{A}|$ (Prop. 5).

A Distributionally Robust Interpretation

The adversary’s KL term makes (5) a regularized worst case rather than a hard one, and the inner problem has a closed form.

Proposition 1 (Soft worst-case form). *For fixed π , let $c_\pi(k, a) = \mathbb{E}_{y \sim \pi} [P_k(y \succ a)]$. Then*

$$\begin{aligned} \min_s V(\pi, s) &= -\kappa \log \mathbb{E}_{(k,a) \sim s_0} \left[\exp \left(-\frac{c_\pi(k, a)}{\kappa} \right) \right] \\ &\quad - \tau \text{KL}(\pi \| \mu), \end{aligned} \quad (7)$$

and, writing $\text{softmin}_\kappa(c_\pi)$ for the first term and $s_{0,\min} = \min_{k,a} s_0(k, a)$,

$$\min_{k,a} c_\pi \leq \text{softmin}_\kappa(c_\pi) \leq \min_{k,a} c_\pi + \kappa \log(1/s_{0,\min}), \quad (8)$$

so $\text{softmin}_\kappa(c_\pi) \rightarrow \min_{k,a} c_\pi(k, a)$ as $\kappa \downarrow 0$.

The proof (Appendix A) is the Gibbs variational principle. The direction of the approximation matters: by (8) the soft minimum sits *above* the hard worst case, so the optimized value is not itself a lower bound on $\min_{k,a} c_\pi(k, a)$. The valid certificate is the left inequality read backwards, $\min_{k,a} c_\pi(k, a) \geq \text{softmin}_\kappa(c_\pi) - \kappa \log(1/s_{0,\min})$, and since $s_{0,\min} \leq 1/(K|\mathcal{A}|)$ the slack grows like $\kappa \log(K|\mathcal{A}|)$ under the uniform prior, quantifying how small κ must be as the pair set grows. The penalized inner problem is the Lagrangian form of a KL-DRO objective over pair distributions centered at s_0 (Hu and Hong 2013); we call κ a *temperature* rather than an ambiguity radius, since the radius of the equivalent constrained program depends on π through the dual. The temperature is the single robustness knob: large κ keeps s near the prior and recovers the fixed s_0 -average over pairs, which matches objective scalarization under a fixed opponent distribution exactly when s_0 factorizes accordingly; small κ approaches a hard selector of the currently weakest pair. We use the soft minimum rather than a hard constraint $c_\pi(k, a) \geq \rho$ because a constraint requires a threshold ρ that is known and feasible a priori (rarely available per judge), whereas the soft minimum pushes the worst case up as far as the geometry allows, keeps the problem

an unconstrained saddle so the monotone machinery applies, and exposes the interpolation above.

Proposition 2 (Pair-level vs. weight-only robustness). (i) For every policy π and every fixed opponent distribution $q \in \Delta(\mathcal{A})$, the weight-only objective of scalar max-min methods (Chakraborty et al. 2024) satisfies $\min_{k \in \mathcal{K}} \mathbb{E}_{a \sim q}[c_\pi(k, a)] \geq \min_{(k, a) \in \mathcal{K} \times \mathcal{A}} c_\pi(k, a)$. (ii) The gap can be $\Omega(1)$ under matched regularization: take $K = |\mathcal{A}| = 2$ and a policy with $c_\pi(1, a_1) = c_\pi(2, a_2) = \frac{1}{2} + \gamma$ and $c_\pi(1, a_2) = c_\pi(2, a_1) = \frac{1}{2} - \gamma$, $\gamma \in (0, \frac{1}{2}]$. Under the uniform opponent the weight-only value is $\frac{1}{2}$ for every γ , and its entropic softening equals $\frac{1}{2}$ exactly, whereas $\min_{k, a} c_\pi = \frac{1}{2} - \gamma$ and, by (8), $\text{softmin}_\kappa(c_\pi) \leq \frac{1}{2} - \gamma + \kappa \log 4 < \frac{1}{2}$ whenever $\kappa < \gamma / \log 4$ (Appendix A).

Part (i) alone is immediate, since a minimum over a larger set cannot increase. Part (ii) is the content: a policy can pass every weight-only audit against a fixed opponent while losing badly to one specific judge–response pair, and the separation survives entropic smoothing at matched temperatures. Only the pair-level adversary exposes this failure mode.

Equilibrium Guarantees

The reason to confine all preference oracles to one adversary is structural: the preference coupling then cancels from the monotonicity gap, no matter how much the oracles conflict.

Lemma 1 (Coupling cancels in the monotonicity gap). For any interior $z = (\pi, s)$ and $z' = (\pi', s')$,

$$\begin{aligned} \langle F(z) - F(z'), z - z' \rangle &= \tau \{ \text{KL}(\pi \| \pi') + \text{KL}(\pi' \| \pi) \} \\ &\quad + \kappa \{ \text{KL}(s \| s') + \text{KL}(s' \| s) \} \\ &\geq m \{ D(z \| z') + D(z' \| z) \}, \end{aligned} \quad (9)$$

where $m = \min\{\tau, \kappa\}$ and $D(z \| z') = \text{KL}(\pi \| \pi') + \text{KL}(s \| s')$.

Proof idea. The bilinear block $\langle -As, A^\top \pi \rangle$ is skew-symmetric, so its contribution $\langle A\delta s, \delta \pi \rangle + \langle A^\top \delta \pi, \delta s \rangle$ vanishes identically. This holds for every matrix A , hence for every collection of oracles and every pool. The separable entropic regularizers contribute the symmetrized KL terms. Full details are in Appendix A. $\square$

Theorem 1 (Existence and uniqueness). The game (5) has a unique saddle point $z^* = (\pi^*, s^*)$, and both components have full support.

Existence follows from Brouwer’s theorem applied to the continuous Gibbs best-response maps; uniqueness combines the saddle variational inequality with the strong monotonicity of Lemma 1 (Appendix A). We call π^* the *robust Nash policy*.

Last-Iterate Convergence

We instantiate OMD (39) on the product $Z = \Delta(\mathcal{Y}) \times \Delta(\mathcal{K} \times \mathcal{A})$ with proximal term $D(z \| z_t) = \text{KL}(\pi \| \pi_t) + \text{KL}(s \| s_t)$; since both the feasible set and the regularizer decompose over blocks, one product step equals two per-block steps,

$$z_{t+1} = \arg \min_{z \in Z} \{ \eta_t \langle F(z_t), z \rangle + D(z \| z_t) \}. \quad (10)$$

For $z_t = (\pi_t, s_t)$, define the policy-side adversarial reward and adversary-side cost

$$r_t(y) = \mathbb{E}_{(k, a) \sim s_t} [P_k(y \succ a)] = (As_t)(y), \quad (11)$$

$$c_t(k, a) = \mathbb{E}_{y \sim \pi_t} [P_k(y \succ a)] = (A^\top \pi_t)(k, a). \quad (12)$$

Substituting (6) into (40) gives the closed-form updates

$$\pi_{t+1}(y) \propto \exp\{\eta_t r_t(y)\} \mu(y)^{\eta_t \tau} \pi_t(y)^{1-\eta_t \tau}, \quad (13)$$

$$s_{t+1}(k, a) \propto \exp\{-\eta_t c_t(k, a)\} s_0(k, a)^{\eta_t \kappa} s_t(k, a)^{1-\eta_t \kappa}. \quad (14)$$

The policy upweights responses that beat the current adversarial mixture; the adversary upweights pairs on which the current policy is weakest. Let $\Theta_t = D(z^* \| z_t)$.

Lemma 2 (Verifiable operator bound). Initialize $z_0 = (\mu, s_0)$ and suppose $\eta_t \tau \leq 1$ and $\eta_t \kappa \leq 1$ for all t . Then the iterates of (13)–(14) satisfy $\|F(z_t)\|_{\oplus, *} \leq G$ with $G = 2 + \max\{\tau, \kappa\}$, where $\|g\|_{\oplus, *} = \max\{\|g_\pi\|_\infty, \|g_s\|_\infty\}$.

Theorem 2 (Exact last-iterate rate). Under the conditions of Lemma 2, the exact dynamics satisfy

$$\Theta_{t+1} \leq (1 - m\eta_t)\Theta_t + \eta_t^2 G^2, \quad (15)$$

with $m = \min\{\tau, \kappa\}$. With $\eta_t = 2/(m(t + t_0))$ and t_0 large enough that $\eta_t \tau \leq 1$ and $\eta_t \kappa \leq 1$, the last iterate obeys $\Theta_T \leq M/(T + t_0)$ with $M = \max\{t_0 \Theta_0, 4G^2/m^2\}$, i.e., $\Theta_T = O(G^2/(m^2 T))$.

Proof idea. Add the one-step entropic-OMD descent inequality over the two blocks with comparators (π^*, s^*) ; the saddle variational inequality makes $\langle F(z^*), z_t - z^* \rangle \geq 0$ and Lemma 1 lower-bounds the remainder by $m\Theta_t$; solve the scalar recurrence (Appendix A). $\square$

The $O(1/T)$ rate is not the best known for KL-regularized preference games: EGPO attains last-iterate linear convergence in the single-oracle case via extragradient updates (Zhou, Fazel, and Du 2025), and since Lemma 1 places the lifted game in the same strongly monotone class, an analogous speedup over $\mathcal{K} \times \mathcal{A}$ is plausible. We keep plain mirror descent deliberately: the practical loss of Section regresses onto the closed form of the OMD step, which is what makes the update partition-free for LLM policies, whereas an extragradient step requires a second, extrapolated oracle evaluation per round, doubling scoring cost and breaking the single-regression structure. We view a linear-rate extragradient variant of the lifted game as a natural extension and do not claim rate optimality; empirically, mirror-prox on random lifted instances converges linearly where OMD tracks its $O(1/T)$ bound (Appendix D, Fig. 3b).

We also record that the rate survives Bernoulli feedback under an idealized query model: at round t , sample one pair $(k_t, a_t) \sim s_t$ and set $\hat{r}_t(y) = \xi_t(y)$ with $\xi_t(y) \sim \text{Ber}(P_{k_t}(y \succ a_t))$ for each y ; independently sample $y_t \sim \pi_t$ and set $\hat{c}_t(k, a) = \zeta_t(k, a)$ with $\zeta_t(k, a) \sim \text{Ber}(P_k(y_t \succ a))$; run (13)–(14) with $(\hat{r}_t, \hat{c}_t)$ in place of (r_t, c_t) .

Corollary 1 (Stochastic preference feedback). Under the step-size conditions of Theorem 2, the stochastic dynamics above satisfy $\mathbb{E}[\Theta_T] \leq M/(T + t_0)$ with the same constants:

$\hat{r}_t, \hat{c}_t \in [0, 1]$ coordinatewise, so the bound of Lemma 2 holds pathwise, and $\mathbb{E}[(\hat{r}_t, \hat{c}_t) \mid \mathcal{F}_t] = (r_t, c_t)$, so the recurrence (15) holds in conditional expectation (Appendix A).

These rates certify the idealized tabular dynamics that the practical loss regresses toward; they are not claims about the parametric prompt-conditioned trainer (see Limitations), matching how guarantees are scoped in prior NLHF practice (Zhang et al. 2025). Corollary 1 also consumes $|\mathcal{Y}| + K|\mathcal{A}|$ preference bits per round, one per coordinate; this full-coordinate feedback is unavailable for language generation and is not the estimator the trainer uses. An analysis under sampled-coordinate or bandit feedback, with explicit dependence on pool coverage and the adversary’s minimum sampling probability, is open and would be the more faithful theory for the deployed loss.

From Exact Updates to a Practical Loss

Partition-Free Log-Ratio Regression

Estimating $r_t(y)$ for every response is expensive, and the normalizer in (13) is inconvenient for parametric policies. Both problems disappear in pairwise log-ratios. For a full-support u , let $\Delta_u(y, y') := \log u(y) - \log u(y')$; all exact iterates are full-support because π_0, μ, s_0 are. Taking log-differences of (13) for two responses cancels the normalizer and yields, with

$$h_t(\pi; y, y') := \Delta_\pi(y, y') - \eta_t \tau \Delta_\mu(y, y') - (1 - \eta_t \tau) \Delta_{\pi_t}(y, y'), \quad (16)$$

the exact-update identity

$$h_t(\pi_{t+1}; y, y') = \eta_t (r_t(y) - r_t(y')). \quad (17)$$

The left side contains only log-probability differences, which are directly computable for an LLM policy, and the unknown partition function is gone. The right side is estimated without integrating over s_t : sample one pair $(k, a) \sim s_t$, query the oracle twice, $z_y \sim \text{Ber}(P_k(y \succ a))$ and $z_{y'} \sim \text{Ber}(P_k(y' \succ a))$, and use the signed difference $z_y - z_{y'}$, which is conditionally unbiased for $r_t(y) - r_t(y')$. The idealized population loss is

$$L_t(\pi) = \mathbb{E} \left[(h_t(\pi; y, y') - \eta_t (z_y - z_{y'}))^2 \right], \quad (18)$$

with expectation over $y, y' \sim \pi_t$ and $(k, a) \sim s_t$.

Proposition 3 (Unbiased partition-free regression). *If the policy class contains the exact update (13), then π_{t+1} minimizes (18); if π_t has full support and policies are identified by pairwise log-ratios on that support, the minimizer is unique (Appendix A).*

The Deployed Surrogate

The trainer keeps the log-ratio left-hand side of (17) and makes three substitutions. *First*, oracle queries are replaced by a deterministic plug-in target built from reward-model scores: per-prompt, per-objective scores are min–max normalized over the candidate pool to $\tilde{r}_k(y \mid x)$, and

$$\hat{P}_k(y \succ a \mid x) = \text{sigmoid}(\lambda(\tilde{r}_k(y \mid x) - \tilde{r}_k(a \mid x))), \quad (19)$$

with a fixed preference scale λ . Min–max normalization is pool-dependent: adding or removing a policy changes $\tilde{r}_k$, and together with λ it sets nearly all plug-in margins, so every comparison in this paper freezes the policy pool per table; raw-scale reporting and a λ sensitivity sweep are flagged in Limitations. Two distinct policy responses y, y' are drawn from the empirical pool policy and paired with the signed relative target

$$\hat{z}_{k,a}(y, y' \mid x) = \hat{P}_k(y \succ a \mid x) - \hat{P}_k(y' \succ a \mid x), \quad (20)$$

so the fields need not be preference-ordered. *Second*, the per-pair target $r_t(y) - r_t(y') = \mathbb{E}_{(k,a) \sim s_t} [\hat{z}_{k,a}(y, y' \mid x)]$ is estimated by objective-stratified sampling rather than by a single pair drawn from s_t . Factor the adversary of (14) as $s_t(k, a) = \omega_t(k) q_t(a \mid k)$, where $\omega_t(k) = \sum_a s_t(k, a)$ is the mass on objective k and $q_t(\cdot \mid k)$ is the induced distribution over responses within that objective. The trainer keeps the objective marginals $\omega_t(k)$ exactly and draws one response $a_k \sim q_t(\cdot \mid k)$ per objective, forming the stratified target

$$\tau_t^{\text{os}}(y, y' \mid x) = \sum_{k \in \mathcal{K}} \omega_t(k) \hat{z}_{k,a_k}(y, y' \mid x), \quad (21)$$

so every objective contributes a training signal on every update at a cost of K plug-in evaluations. Over the independent draws a_k , the target is unbiased for the exact adversary expectation, and its variance never exceeds that of a single pair sampled from s_t (Propositions 4 and 5 in Appendix B). The trained loss is

$$\hat{L}_t(\pi) = \mathbb{E} \left[(h_t(\pi; y, y') - \eta_t \tau_t^{\text{os}}(y, y' \mid x))^2 \right], \quad (22)$$

with Δ_{π_t} precomputed under the policy that generated the pool. *Third*, the training pairs (y, y') are not two independent draws from π_t : the deployed pipelines select pairs from the pool deterministically (a best-vs-adversary pairing at 1.5B; three deterministic pairs per prompt on SafeRLHF; Appendix E). Conditional on a selected pair the stratified target remains unbiased for the adversary expectation, but the distribution over pairs differs from the two-draw distribution of Proposition 3, so the population-level exactness and uniqueness arguments do not transfer verbatim to the deployed sampler. Algorithm 1 summarizes the procedure. The stratified target occupies the middle of a cost–fidelity spectrum. Appendix B defines a full-expectation target that keeps every atom of s_t and a cheaper top-mass target that keeps only the heaviest atoms, proves the unbiasedness and variance-reduction properties above, and reports an estimator ablation in which the stratified target attains the best average and worst-objective reward.

Remark 1 (What the surrogate inherits). *Two properties of Prop. 3 survive the substitutions and two do not. Survive: the regression remains partition-free, and the stratified target is conditionally unbiased for the adversary expectation $\mathbb{E}_{(k,a) \sim s_t} [\hat{z}_{k,a}]$ of the plug-in preferences, so the regression still points toward the OMD step of the surrogate oracle. Do not survive: the adversary weights are computed from cost estimates $\hat{c}_t$ over the finite pool, so the effective adversary*

Algorithm 1 RONPO for prompt-conditioned policies

- 1: Initialize $\pi_0(\cdot | x) = \mu(\cdot | x)$ and adversary prior $s_0(\cdot | x)$ over $\mathcal{K} \times \mathcal{A}(x)$.
 - 2: **for** $t = 0, 1, \dots, T - 1$ **do**
 - 3: Build or refresh the response pool $\mathcal{A}_t(x)$ (reference, current, and historical policies; external teachers).
 - 4: Score $\mathcal{A}_t(x)$ with all K objectives; normalize per prompt-objective; form $\hat{P}_k$ by (19).
 - 5: Estimate $\hat{c}_t(k, a | x) = \mathbb{E}_{y \sim \hat{\pi}_t}[\hat{P}_k(y \succ a | x)]$ over the pool.
 - 6: Form the adversary $s_t(k, a) \propto s_0(k, a) \exp\{-\hat{c}_t(k, a)/\kappa\}$, the entropic best response (31) (one step of (14) at $\eta = 1/\kappa$); factor $s_t(k, a) = \omega_t(k) q_t(a | k)$ and draw one opponent $a_k \sim q_t(\cdot | k)$ per objective.
 - 7: Select a training pair (y, y') from the pool (two independent draws from $\hat{\pi}_t(\cdot | x)$ in the analyzed dynamics; deterministic selection in the deployed pipelines); form the stratified target τ_t^{ps} by (21) from the per-pair values (20).
 - 8: Precompute $\Delta_\mu(y, y')$ and $\Delta_{\pi_t}(y, y')$; update the policy by minimizing $\hat{L}_t$ in (22).
 - 9: **end for**
 - 10: **return** π_T .
-

is exact only relative to that pool; $\hat{P}_k$ carries plug-in bias governed by λ ; and the pair distribution is the deterministic selection above rather than two independent policy draws. Moreover, because (19) is logistic in a scalar score gap, the plug-in oracle is Bradley–Terry and hence transitive: at model scale RONPO exercises heterogeneous-reward robustness, and genuine intransitivity is exercised only in the synthetic games, whose oracles are arbitrary skew-symmetric matrices. We phrase all model-scale claims accordingly.

Adversary schedule. The analyzed dynamics update the adversary by the coupled mirror step (14). The implementation instead recomputes the adversary from the current cost estimate at every policy stage. This computes best response $s_t(k, a) \propto s_0(k, a) \exp\{-\hat{c}_t(k, a)/\kappa\}$ of (31), which coincides with a single step of (14) at the largest admissible step size $\eta = 1/\kappa$. Theorem 2 analyzes the simultaneous coupled dynamics, not this stagewise best-response schedule, so at model scale the theorem motivates the update rather than certifying the deployed dynamics.

Experiments

Our evaluation has two parts. Synthetic preference games (i) isolate the hidden-objective failure of averaging in a setting where the robust optimum is computable exactly, and (ii) numerically validate Theorems 1–2, Corollary 1, and Proposition 1. LLM experiments then test whether the deployed surrogate (Sec.) delivers heterogeneous-reward robustness at model scale.

Because worst-case language is easy to overload, we fix names for three distinct quantities: the theory’s *pairwise floor* $\mathbb{E}_x[\min_{k,a} P_k(\pi \succ a | x)]$; the *worst objective mean*

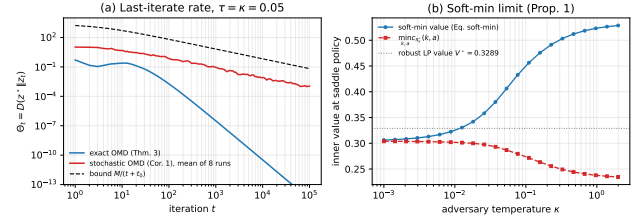


Figure 1: Numerical validation of the theory on the Table 5 instance. (a) Last-iterate Lyapunov value $\Theta_t = D(z^* || z_t)$ under the step schedule of Theorem 2 ($\tau = \kappa = 0.05$). Exact dynamics converge below the explicit bound $M/(t + t_0)$; with single-pair Bernoulli feedback (Cor. 1, mean of 8 runs) the decay is $\Theta(1/t)$ (empirical log–log slope -0.90 over the final decade), running parallel to the bound. (b) As $\kappa \downarrow 0$ the inner soft-min value and the hard minimum $\min_{k,a} c_{\pi_k^*}(k, a)$ at the saddle policy converge *together*, as the sandwich bound (8) requires; their common limit is the τ -regularized robust optimum, which sits slightly below the unregularized LP value $V^* = 0.3289$ (dotted) because $\tau = 0.05$ is held fixed. Recovering V^* exactly would require the joint limit $\tau, \kappa \downarrow 0$ (Prop. 1).

$\min_k \mathbb{E}_x[\tilde{r}_k(x)]$, the minimum over objectives of the per-objective mean reward, used by the 1.5B tables; and the *per-prompt worst* $\mathbb{E}_x[\min_k \tilde{r}_k(x)]$, the mean of the per-prompt minimum, used by the SafeRLHF table. The two reward aggregates drop the opponent-response dimension, and exchanging the minimum and the expectation can reorder methods, so each table states which it reports; the win rates WR_B/wWR_B are pairwise but fix the base response as the single opponent. Estimating the pairwise floor itself against a rich frozen response pool remains the main gap between the theory’s endpoint and these metrics.

Numerical Validation on Synthetic Games

Setup. We validate the convergence theory on a synthetic decoy instance with $|\mathcal{Y}| = 8$ actions and $K = 3$ conflicting objectives. The full game construction, the compared tabular methods, and the averaging-failure comparison that motivates the robust objective are deferred to Appendix D; the robust linear-programming value on this instance is $V^* = 0.3289$.

The dynamics track the predicted rates. Figure 1 ties the numerics to the statements. Panel (a) plots Θ_t against the explicit bound $M/(t + t_0)$ of Theorem 2: the exact dynamics stay below the bound at every iterate (converging faster, as expected for a deterministic map whose fixed point is z^*), while stochastic Bernoulli feedback saturates the $\Theta(1/t)$ rate of Corollary 1. Panel (b) verifies the soft-min characterization: as $\kappa \downarrow 0$ the regularized inner value and the hard pair minimum at the saddle policy converge together, confirming the closed form (7) and the sandwich (8); their joint limit is the τ -regularized robust value, visibly below V^* because τ stays fixed, so the panel quantifies the fixed- τ gap rather than claiming convergence to V^* itself. We stress the scope

of this subsection: it verifies the algebraic statements on one designed instance, as a consistency check of the theory, not as a general benchmark of the optimizer. Appendix D complements it with an empirical study over random game families: confidence bands across games and seeds, an extragradient comparison, and sensitivity to $|\mathcal{Y}|$, K , pool coverage, $\tau = \kappa$, and payoff noise. All curves are produced by a self-contained script included with the supplementary code.

LLM Alignment Experiments

Setup. All policies are initialized from Qwen2.5-1.5B-Instruct and trained on 19,856 UltraFeedback prompts with three heterogeneous reward objectives (Skywork, Athene, and ArmoRM; Appendix E), applied through the plug-in oracle (19). Training proceeds in two stages of iterative pool construction and optimization; the response pool $\mathcal{A}(x)$ mixes reference-, current-, and historical-policy generations. Evaluation uses 647 held-out prompts and three axes: (i) the three *local reward models* (training-aligned, so reported as an in-distribution robustness probe: per-prompt normalized reward, the worst objective mean, and win rate against the base response); (ii) *IFEval* (Zhou et al. 2023), testing whether preference optimization damages verifiable instruction following; and (iii) a reward-model-independent gpt-5.5 all-pairwise judge with deterministically randomized order, ties counted as 0.5, and prompt-level bootstrap confidence intervals (protocol in Appendix E).

Baselines. We compare against (a) the base policy; (b) three single-oracle HT-MNPO policies, each trained on one reward model, which represent the “pick one judge” strategy; and (c) averaged-oracle self-play baselines SPPO-avg and INPO-avg trained on the prompt-wise average of the min-max-normalized objectives. All trained methods share the same base policy, data, and response pools. The averaged-oracle baselines are evaluated at stage 1; eligible stage-2 continuations of SPPO-avg and INPO-avg are not available, so all stage-2 comparisons are against the base and single-oracle policies, and every claim about fixed averaging in this study rests on the stage-1 comparison. We also note two protocol gaps on the robust-baseline axis: no dedicated worst-objective trainer (MaxMin-RLHF, group-robust preference optimization, or MOPO) is included, and the factorized ablation that adversarially selects only the objective k while holding the opponent distribution fixed, the most direct test of the joint (k, a) index, is left to future work.

Stage-1 results. On the first-stage checkpoints (full tables in Appendix E), RONPO obtains the best average and worst-objective normalized reward (0.637/0.633 versus 0.500/0.467 for the strongest single-oracle baseline) and by a wide margin the smallest cross-objective spread (0.005), so its advantage is spread across the three reward models rather than resting on one. The SPPO-avg and INPO-avg rows are repaired baselines, retrained with a stability-gated hyperparameter sweep so that the comparison is against properly trained averaging and not a failed run. Once repaired, INPO-avg is a strong baseline and edges RONPO on the Athene objective and on the worst-objective win rate, but RONPO

Method	Local RM				IFEval	
	Avg	Worst	WR _B	wWR _B	Strict	Tok.
Base	0.4852	0.3082	–	–	0.4140	610
HT-MNPO (Sky.)	<u>0.5020</u>	<u>0.3428</u>	<u>52.29</u>	<u>47.53</u>	0.3530	1166
HT-MNPO (Ath.)	0.3881	0.2377	43.84	38.02	0.3290	1970
HT-MNPO (Armo)	0.4148	0.2553	45.36	37.87	0.3216	1571
RONPO	0.7025	0.5475	66.05	60.05	<u>0.3993</u>	725

Table 2: Stage-2 summary. Local RM: per-prompt min–max-normalized reward, reporting the mean of the three objective scores (Avg), the per-prompt worst objective averaged over prompts $\mathbb{E}_x[\min_k \tilde{r}_k]$ (Worst), and average/worst win rate vs. the base response (WR_B/wWR_B, %). IFEval: prompt-strict accuracy and mean output tokens. RONPO improves every robustness metric while staying far closer to the base model’s instruction following and length than the single-oracle baselines.

keeps the highest average and worst-objective reward and the most even objective profile. We read this narrowly: even a well-trained fixed-average baseline leaves the weakest objective lower than the robust objective does.

Stage-2 results. Table 2 summarizes the final stage. RONPO reaches 0.703/0.670 average/worst normalized reward versus 0.502/0.474 for the strongest single-oracle baseline, with a 66.0% average win rate over the base response. On IFEval the base model remains best, so this axis measures *preservation*: RONPO stays near the base prompt-strict score with moderate output length, while the single-oracle baselines degrade more and generate much longer outputs, so the reward gains are not a length artifact. On the reward-model-independent GPT-5.5 all-pairwise tournament (Appendix E), which covers the base, the three single-oracle policies, and RONPO but not the averaged baselines, RONPO ranks first overall, beating every included trained baseline (58.7–68.6% head-to-head) while reaching only parity with the base policy (51.8%, CI overlapping 0.5). The honest summary is narrower than “transfers favorably”: under the training-aligned reward models RONPO improves every robustness metric, and under the external judge it beats the included trained baselines but shows no significant gain over the base policy.

Model-scale robustness on SafeRLHF. We test whether the deployed surrogate delivers heterogeneous-reward robustness at the 8B scale on a task with a genuine helpfulness–harmlessness conflict. The primary block initializes all policies from Llama-3.1-8B-Instruct and trains them on PKU-SafeRLHF (Dai et al. 2024) for four iterative stages of pool construction and optimization (Appendix E.3), applying the plug-in oracle (19) to two objectives: helpfulness, scored by the Beaver reward model, and harmlessness, scored by the negated Beaver cost model. All methods share the base policy, the training prompts, and the response pools. We evaluate on a held-out panel of 1,000 SafeRLHF prompts that is disjoint from training, rescore every response with both Beaver heads, and normalize each objective per prompt

over the compared policies; every eligible policy passes the generation stability gate.

Results. Table 3 reports the mean and sample standard deviation across training seeds 42, 43, and 44 for per-prompt normalized helpfulness and harmlessness, their per-prompt average, and the per-prompt worst $\mathbb{E}_x[\min_k \tilde{r}_k]$ (the naming convention above); the $\pm$ values describe seed-level spread and are not confidence intervals on method differences. RONPO gives the best worst-objective reward (0.4408 ± 0.0061) and the best average (0.6231 ± 0.0048), ahead of every baseline and far above the base policy. The gain comes from balance: RONPO is strong on both objectives (0.6578 ± 0.0111 helpfulness and 0.5883 ± 0.0048 harmlessness), whereas methods that spike a single objective, such as SimPO with 0.7431 ± 0.0089 harmlessness at 0.4672 ± 0.0118 helpfulness, pay for it on the worst objective. IPO is the strongest baseline and trails RONPO by 0.0274 on the three-seed mean of the per-prompt worst. RONPO leads IPO in each seed individually, which is the comparison we weight, and we attach no significance claim to the three-seed mean difference; with three seeds we read this experiment as consistently favorable but exploratory. INPO-avg is the strongest averaging baseline but still trails RONPO and IPO on the worst objective, while the single-oracle HT-MNPO policies and SPPO-avg are the weakest trained policies there. The Qwen2.5-7B replication uses the same three seeds and a frozen common pool of Stage-4 gate passers. RONPO again leads on Worst (0.7073 ± 0.0124) and Avg (0.8258 ± 0.0112), while SimPO is the strongest eligible baseline on both aggregates. All listed Qwen methods have a valid Stage-4 checkpoint in every seed. Because each backbone is normalized over its own frozen policy pool, values are not compared across blocks.

Conclusion

RONPO moves the objective index into a single adversary’s action space, keeping heterogeneous preference alignment inside a KL-regularized two-player bilinear saddle. On a finite oracle–response pool the game has a unique robust Nash policy, and exact and stochastically fed tabular mirror-descent dynamics converge to it at an $O(1/T)$ last-iterate rate. The inner problem is an entropy-smoothed conservative objective over judge–response pairs, an upper approximation of the hard worst case controlled to within $\kappa \log(1/s_{0,\min})$, and the idealized update is a partition-free log-ratio regression. Synthetic games confirm the closed forms and rates on a designed instance and show that fixed averaging can hide a minority-objective collapse the pair adversary exposes. At model scale, where the oracles are scalar reward models behind a Bradley–Terry plug-in, RONPO improves worst-objective and average reward under the training-aligned judges, consistently across the three SafeRLHF seeds, while remaining statistically tied with the base policy under an external judge; benefits under genuinely non-transitive preferences at model scale remain untested.

Limitations. The convergence guarantees are exact for the tabular per-context dynamics and their stochastically fed

Method	Helpful.	Harmless.	Avg	Worst
<i>Base model: Llama-3.1-8B-Instruct</i>				
RONPO	0.6578±0.0111	0.5883±0.0048	0.6231±0.0048	0.4408±0.0061
HT-MNPO (help.)	0.5263±0.0166	0.4301±0.0103	0.4782±0.0046	0.2932±0.0057
HT-MNPO (harm.)	0.5185±0.0132	0.4335±0.0042	0.4760±0.0084	0.2920±0.0126
INPO-avg	0.4932±0.0317	0.6413±0.0096	0.5673±0.0149	0.3732±0.0192
SPPO-avg	0.4834±0.0042	0.4697±0.0127	0.4765±0.0073	0.2844±0.0139
SimPO	0.4672±0.0118	0.7431±0.0089	0.6051±0.0081	0.3783±0.0092
IPO	0.5614±0.0340	0.6468±0.0123	0.6041±0.0109	0.4134±0.0153
DPO	0.4971±0.0100	<u>0.6560±0.0120</u>	0.5766±0.0083	0.3784±0.0111
Base	0.2019±0.0047	0.4514±0.0100	0.3266±0.0030	0.1310±0.0011
<i>Base model: Qwen2.5-7B-Instruct</i>				
RONPO	0.8092±0.0334	0.8425±0.0515	0.8258±0.0112	0.7073±0.0124
HT-MNPO (help.)	0.5604±0.0644	0.4017±0.0222	0.4810±0.0418	0.2959±0.0441
HT-MNPO (harm.)	0.4530±0.0472	0.5096±0.0335	0.4813±0.0399	0.2986±0.0433
INPO-avg	0.3426±0.0905	0.4738±0.0422	0.4082±0.0658	0.2259±0.0672
SPPO-avg	0.5369±0.1422	0.4219±0.0740	0.4794±0.0349	0.2807±0.0070
SimPO	0.6895±0.0805	0.6947±0.0367	0.6921±0.0222	0.5231±0.0251
IPO	0.3291±0.0707	0.4847±0.0471	0.4069±0.0581	0.2179±0.0545
DPO	0.5185±0.1202	0.5608±0.1063	0.5397±0.0811	0.3628±0.0788
Base	0.2469±0.0126	0.4880±0.0553	0.3675±0.0311	0.1664±0.0172

Table 3: **Model-scale robustness on PKU-SafeRLHF.** Mean $\pm$ sample standard deviation across independent training runs with 3 seeds. Helpfulness (Beaver reward) and harmlessness (negated Beaver cost) are min–max normalized per prompt over a frozen eligible policy pool for each backbone; Avg and Worst are the per-prompt mean and per-prompt minimum of the two normalized objectives averaged over the held-out 1,000-prompt panel, so Worst is the per-prompt worst $\mathbb{E}_x[\min_k \tilde{r}_k]$. Best in bold and second best underlined within each backbone.

variant; the prompt-conditioned trainer uses a shared network, a stratified adversary target with one sampled opponent per objective, deterministic pair selection, a stagewise best-response adversary schedule (Sec.), and the plug-in oracle (19), and its robustness is relative to the finite pool $\mathcal{A}$ and the K oracles. All reward-based numbers use per-prompt min–max normalization within a frozen policy pool, so rankings are internally consistent but not comparable across pools; raw-scale reporting, set-independent calibration, and a λ sensitivity sweep remain open. Because the plug-in oracle is Bradley–Terry, model-scale claims concern heterogeneous-reward robustness; intransitivity is exercised only in the synthetic games. Most model-scale runs use a single training seed. The reward and external-judge study uses a 1.5B backbone with prompt-level (not seed-level) confidence intervals, and its local reward judges overlap the training objectives; the external judge is the only reward-model-independent signal and shows parity with the base policy on the hardest pairwise subset. The 8B SafeRLHF result uses three training seeds and gives RONPO the best worst-objective and average reward in all three, but three seeds still provide only a coarse estimate of between-seed uncertainty. A separate Qwen3-8B diagnostic with three ArmoRM objectives is underpowered, since its raw head means vary by at most 0.0045 across methods, so that normalized ranking (Appendix E.5) reflects evaluator power rather than model robustness. Additional seeds and reward-independent judges are required before a stronger claim can be made.

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A Deferred Proofs

Throughout, $z = (\pi, s)$ and $z' = (\pi', s')$ denote interior points of $Z = \Delta(\mathcal{Y}) \times \Delta(\mathcal{K} \times \mathcal{A})$, and $\delta\pi := \pi - \pi'$, $\delta s := s - s'$.

A.1 Proof of Lemma 1 (Monotonicity Gap)

Because z and z' are interior, every coordinate of π, π', s, s' is strictly positive, so $\log \pi, \log \pi', \log s, \log s'$ are finite, the operator F of (6) is well defined at both points, and every

Kullback–Leibler term below is finite. Equip $Z \subseteq \mathbb{R}^{|\mathcal{Y}|} \times \mathbb{R}^{K|\mathcal{A}|}$ with the product inner product

$$\langle (u_1, v_1), (u_2, v_2) \rangle = \langle u_1, u_2 \rangle + \langle v_1, v_2 \rangle. \quad (23)$$

Step 1 (Additive split of the operator). By (6), decompose $F = B + g$ with

$$\begin{aligned} B(\pi, s) &:= (-As, A^\top \pi), \\ g(\pi, s) &:= (\tau(\log(\pi/\mu) + \mathbf{1}), \kappa(\log(s/s_0) + \mathbf{1})). \end{aligned}$$

The map B is linear on Z with block representation

$$B = \begin{pmatrix} 0 & -A \\ A^\top & 0 \end{pmatrix}. \quad (24)$$

Subtracting $F(z')$ from $F(z)$ and using linearity of B ,

$$F(z) - F(z') = B(\delta\pi, \delta s) + (g(z) - g(z')). \quad (25)$$

In $g(z) - g(z')$, the reference vectors $-\tau \log \mu, -\kappa \log s_0$ and the all-ones vectors cancel, leaving

$$g(z) - g(z') = (\tau(\log \pi - \log \pi'), \kappa(\log s - \log s')). \quad (26)$$

Step 2 (The bilinear coupling cancels). By (24) and (23),

$$\langle B(\delta\pi, \delta s), (\delta\pi, \delta s) \rangle = -\langle A \delta s, \delta\pi \rangle + \langle A^\top \delta\pi, \delta s \rangle.$$

Expanding entrywise, with $A_{y,(k,a)} = P_k(y \succ a)$,

$$\langle A \delta s, \delta\pi \rangle = \sum_y \sum_{k,a} A_{y,(k,a)} \delta\pi_y \delta s_{k,a} = \langle A^\top \delta\pi, \delta s \rangle,$$

the same finite double sum in a different order (the adjoint property of the transpose). Carrying opposite signs, the two terms cancel:

$$\langle B(\delta\pi, \delta s), (\delta\pi, \delta s) \rangle = 0. \quad (27)$$

Nothing about A was used beyond its being a fixed matrix, so the coupling vanishes for every collection of oracles $P_1, \dots, P_K$ and every pool $\mathcal{A}$; structurally, (27) says B is skew-symmetric, $B^\top = -B$.

Step 3 (The regularizer yields a symmetrized KL). Pairing (26) against $(\delta\pi, \delta s)$,

$$\begin{aligned} \langle g(z) - g(z'), (\delta\pi, \delta s) \rangle &= \tau \langle \log \pi - \log \pi', \pi - \pi' \rangle \\ &\quad + \kappa \langle \log s - \log s', s - s' \rangle. \end{aligned} \quad (28)$$

Splitting the policy-block sum into its $\pi(y)$ - and $\pi'(y)$ -weighted parts,

$$\begin{aligned} \langle \log \pi - \log \pi', \pi - \pi' \rangle &= \sum_y \pi(y) (\log \pi(y) - \log \pi'(y)) \\ &\quad + \sum_y \pi'(y) (\log \pi'(y) - \log \pi(y)) \\ &= \text{KL}(\pi \| \pi') + \text{KL}(\pi' \| \pi), \end{aligned}$$

and identically for the adversary block. Substituting into (28),

$$\begin{aligned} \langle g(z) - g(z'), (\delta\pi, \delta s) \rangle &= \tau \{ \text{KL}(\pi \| \pi') + \text{KL}(\pi' \| \pi) \} \\ &\quad + \kappa \{ \text{KL}(s \| s') + \text{KL}(s' \| s) \}. \end{aligned} \quad (29)$$

Step 4 (Assembling). Pairing (25) against $z - z'$ and using (27) and (29) gives (9).

Step 5 (Strong-monotonicity bound). Relative entropy is nonnegative, so replacing τ and κ by $m = \min\{\tau, \kappa\}$ in front of nonnegative terms can only decrease the value:

$$\langle F(z) - F(z'), z - z' \rangle \geq m\{D(z\|z') + D(z'\|z)\},$$

with $D(z\|z') = \text{KL}(\pi\|\pi') + \text{KL}(s\|s')$. $\square$

A.2 Proof of Theorem 1 (Existence and Uniqueness)

For fixed s , $\pi \mapsto V(\pi, s)$ is strictly concave (linear minus $\tau \text{KL}(\pi\|\mu)$), with unique best response

$$B_\pi(s)(y) = \frac{\mu(y) \exp((As)_y/\tau)}{\sum_u \mu(u) \exp((As)_u/\tau)}. \quad (30)$$

For fixed π , $s \mapsto V(\pi, s)$ is strictly convex, with unique best response

$$B_s(\pi)(k, a) = \frac{s_0(k, a) \exp(-(A^\top \pi)_{k,a}/\kappa)}{\sum_{j,b} s_0(j, b) \exp(-(A^\top \pi)_{j,b}/\kappa)}. \quad (31)$$

The map $(\pi, s) \mapsto (B_\pi(s), B_s(\pi))$ is continuous on the compact convex product of simplices, so Brouwer's theorem gives a fixed point (π^*, s^*) with $\pi^* = B_\pi(s^*)$ and $s^* = B_s(\pi^*)$. Since π^* is the exact policy best response to s^* and s^* the exact adversary best response to π^* ,

$$V(\pi, s^*) \leq V(\pi^*, s^*) \leq V(\pi^*, s) \quad \forall \pi, s, \quad (32)$$

which is the definition of a saddle point. Every saddle point must satisfy the same best-response equations, and since μ, s_0 are strictly positive and the exponentials in (30)–(31) are strictly positive, every saddle point has full support.

For uniqueness, we first derive the saddle variational inequality. Fix any feasible π and set $\pi_\epsilon := \pi^* + \epsilon(\pi - \pi^*)$ for $\epsilon \in [0, 1]$; feasibility follows from convexity of the simplex. Since π^* maximizes $V(\cdot, s^*)$,

$$\frac{V(\pi_\epsilon, s^*) - V(\pi^*, s^*)}{\epsilon} \leq 0 \implies \langle \nabla_\pi V(z^*), \pi - \pi^* \rangle \leq 0$$

by letting $\epsilon \downarrow 0$ and using differentiability at the interior point z^* . Likewise, minimality of s^* for $V(\pi^*, \cdot)$ gives $\langle \nabla_s V(z^*), s - s^* \rangle \geq 0$ for all feasible s . Negating the first inequality, adding the second, and using $F = (-\nabla_\pi V, \nabla_s V)$ yields

$$\langle F(z^*), z - z^* \rangle \geq 0 \quad \forall z \in Z. \quad (33)$$

Now let z_1, z_2 be two saddle points. Applying (33) at z_1 with $z = z_2$ and at z_2 with $z = z_1$, then adding,

$$\langle F(z_1) - F(z_2), z_1 - z_2 \rangle \leq 0.$$

Lemma 1 lower-bounds the same quantity by $m\{D(z_1\|z_2) + D(z_2\|z_1)\} \geq 0$. Both hold only if the divergences vanish, hence $z_1 = z_2$. $\square$

A.3 Proof of Propositions 1 and 2

For fixed π , $\min_s V(\pi, s) = -\tau \text{KL}(\pi\|\mu) + \min_s \{\langle c_\pi, s \rangle + \kappa \text{KL}(s\|s_0)\}$. The inner minimization is the Gibbs variational principle: for any cost vector c and prior s_0 ,

$$\min_{s \in \Delta} \{\langle c, s \rangle + \kappa \text{KL}(s\|s_0)\} = -\kappa \log \mathbb{E}_{s_0}[e^{-c/\kappa}],$$

attained uniquely at $s^* \propto s_0 \exp(-c/\kappa)$, obtained by setting the gradient of the strictly convex objective to a constant on the simplex. Substituting $c = c_\pi$ gives (7).

For the sandwich (8), write $c^* = \min_i c_i$. Since $e^{-c_i/\kappa} \leq e^{-c^*/\kappa}$ for every i and s_0 is a distribution, $\mathbb{E}_{s_0}[e^{-c/\kappa}] \leq e^{-c^*/\kappa}$, and applying the decreasing map $-\kappa \log(\cdot)$ gives $\text{softmax}_\kappa(c) \geq c^*$. Conversely, keeping only an atom $i^* \in \arg \min_i c_i$, $\mathbb{E}_{s_0}[e^{-c/\kappa}] \geq s_0(i^*)e^{-c^*/\kappa} \geq s_{0,\min}e^{-c^*/\kappa}$, so $\text{softmax}_\kappa(c) \leq c^* + \kappa \log(1/s_{0,\min})$. The two inequalities pinch as $\kappa \downarrow 0$, which recovers the log-sum-exp limit $\text{softmax}_\kappa(c) \rightarrow c^*$ for full-support s_0 .

For Proposition 2(i): for any $k \in \mathcal{K}$ and $q \in \Delta(\mathcal{A})$, $\mathbb{E}_{a \sim q}[c_\pi(k, a)] \geq \min_{a \in \mathcal{A}} c_\pi(k, a) \geq \min_{(k', a) \in \mathcal{K} \times \mathcal{A}} c_\pi(k', a)$, since an average dominates a minimum and a minimum over a larger set is no larger; taking the minimum over k preserves the inequality. For (ii): under the stated costs, $\mathbb{E}_{a \sim \text{unif}}[c_\pi(k, a)] = \frac{1}{2}$ for both k , so the weight-only value is $\frac{1}{2}$, and its entropic softening over k is $-\kappa \log(\frac{1}{2}e^{-1/(2\kappa)} + \frac{1}{2}e^{-1/(2\kappa)}) = \frac{1}{2}$ because the two coordinates coincide. The joint minimum is $\frac{1}{2} - \gamma$, and (8) with the uniform prior over the four pairs ($s_{0,\min} = \frac{1}{4}$) gives $\text{softmax}_\kappa(c_\pi) \leq \frac{1}{2} - \gamma + \kappa \log 4$, strictly below $\frac{1}{2}$ whenever $\kappa < \gamma/\log 4$. $\square$

A.4 Proof of Lemma 2 (Operator Bound)

We bound the policy block $F_\pi(z_t) = -As_t + \tau(\psi_t + \mathbf{1})$ with $\psi_t := \log(\pi_t/\mu)$; the adversary block is symmetric. Since $P_k \in [0, 1]$ and s_t is a distribution, $(As_t)_y = \mathbb{E}_{(k,a) \sim s_t}[P_k(y \succ a)] \in [0, 1]$, so $\|As_t\|_\infty \leq 1$.

Normalization. Because π_t and μ are distributions, $\mathbb{E}_\mu[e^{\psi_t}] = \sum_y \mu(y) \pi_t(y)/\mu(y) = 1$, so ψ_t is neither everywhere positive nor everywhere negative: $\min_y \psi_t(y) \leq 0 \leq \max_y \psi_t(y)$.

Spread. Taking logs of (13), $\psi_{t+1}(y) = \eta_t r_t(y) + (1 - \eta_t \tau) \psi_t(y) - \log Z_t$ with normalizer Z_t ; the constant cancels in differences, so for all y, y' , using $1 - \eta_t \tau \geq 0$,

$$\begin{aligned} \psi_{t+1}(y) - \psi_{t+1}(y') &= \eta_t(r_t(y) - r_t(y')) \\ &\quad + (1 - \eta_t \tau)(\psi_t(y) - \psi_t(y')). \end{aligned}$$

Let $\sigma_t := \max_y \psi_t(y) - \min_y \psi_t(y)$. As $r_t(y) \in [0, 1]$, its spread is at most 1, so $\sigma_{t+1} \leq \eta_t + (1 - \eta_t \tau)\sigma_t$. From $\pi_0 = \mu$ (so $\sigma_0 = 0$), induction yields $\sigma_t \leq 1/\tau$: if $\sigma_t \leq 1/\tau$ then $\sigma_{t+1} \leq \eta_t + (1 - \eta_t \tau)/\tau = 1/\tau$.

Combine. From $\min_y \psi_t \leq 0 \leq \max_y \psi_t$ and $\sigma_t \leq 1/\tau$ we get $\|\psi_t\|_\infty \leq 1/\tau$, hence

$$\begin{aligned} \|F_\pi(z_t)\|_\infty &\leq \|As_t\|_\infty + \tau\|\psi_t + \mathbf{1}\|_\infty \\ &\leq 1 + \tau(\frac{1}{\tau} + 1) = 2 + \tau. \end{aligned}$$

The identical argument for $\log(s_t/s_0)$ gives $\|F_s(z_t)\|_\infty \leq 2 + \kappa$; taking the maximum gives $G = 2 + \max\{\tau, \kappa\}$. Note

that only $r_t \in [0, 1]^{|Y|}$ and $c_t \in [0, 1]^{K|A|}$ coordinatewise were used, a fact we reuse in Corollary 1. $\square$

A.5 Proof of Theorem 2 (Last-Iterate Rate)

Apply the descent form of the one-step entropic-OMD inequality (47) separately to the two blocks: policy block with $u_t = \pi_t$, comparator π^* , gradient $F_\pi(z_t)$; adversary block with $u_t = s_t$, comparator s^* , gradient $F_s(z_t)$. Adding,

$$\begin{aligned} \Theta_{t+1} &\leq \Theta_t - \eta_t \langle F(z_t), z_t - z^* \rangle \\ &\quad + \frac{\eta_t^2}{2} (\|F_\pi(z_t)\|_\infty^2 + \|F_s(z_t)\|_\infty^2), \end{aligned} \quad (34)$$

where the two middle terms combined via the product inner product. Lemma 2 bounds the last term by $\eta_t^2 G^2$. Decompose the inner product as $\langle F(z_t) - F(z^*), z_t - z^* \rangle + \langle F(z^*), z_t - z^* \rangle$: the variational inequality (33) makes the second term nonnegative, and Lemma 1 bounds the first below by $m\{D(z_t \| z^*) + D(z^* \| z_t)\} \geq m\Theta_t$. This gives (15).

It remains to solve the recurrence. Set $a_t := \Theta_t$ and $C := 4G^2/m^2$. With $\eta_t = 2/(m(t + t_0))$, (15) reads

$$a_{t+1} \leq \left(1 - \frac{2}{t + t_0}\right) a_t + \frac{C}{(t + t_0)^2}.$$

Let $M := \max\{t_0 a_0, C\}$. We show $a_t \leq M/(t + t_0)$ by induction: the base case is $a_0 \leq M/t_0$; for the step, write $n = t + t_0$, so

$$\begin{aligned} a_{t+1} &\leq \left(1 - \frac{2}{n}\right) \frac{M}{n} + \frac{C}{n^2} = \frac{M(n-2) + C}{n^2} \\ &\leq \frac{M(n-1)}{n^2} \leq \frac{M}{n+1}, \end{aligned}$$

using $C \leq M$ and $(n-1)(n+1) \leq n^2$. $\square$

A.6 Proof of Corollary 1 (Stochastic Feedback)

Let $\mathcal{F}_t$ be the σ -field generated by the history through z_t . The realized gradients are $\hat{F}_\pi(z_t) = -\hat{r}_t + \tau(\psi_t + \mathbf{1})$ and $\hat{F}_s(z_t) = \hat{c}_t + \kappa(\log(s_t/s_0) + \mathbf{1})$.

Pathwise bound. By construction $\hat{r}_t(y) = \xi_t(y) \in \{0, 1\} \subset [0, 1]$ and $\hat{c}_t(k, a) = \zeta_t(k, a) \in \{0, 1\} \subset [0, 1]$ coordinatewise. The spread induction in the proof of Lemma 2 used only that the reward vector lies in $[0, 1]$ coordinatewise, so it applies verbatim to the stochastic iterates and yields, pathwise, $\|\hat{F}_\pi(z_t)\|_\infty \leq G$ and $\|\hat{F}_s(z_t)\|_\infty \leq G$.

Unbiasedness. Conditional on $\mathcal{F}_t$, $\mathbb{E}[\xi_t(y)] = \mathbb{E}_{(k,a) \sim s_t}[P_k(y \succ a)] = r_t(y)$ for every y , and $\mathbb{E}[\zeta_t(k, a)] = \mathbb{E}_{y \sim \pi_t}[P_k(y \succ a)] = c_t(k, a)$ for every (k, a) ; hence $\mathbb{E}[\hat{F}(z_t) | \mathcal{F}_t] = F(z_t)$, since the regularizer terms are $\mathcal{F}_t$ -measurable.

Recurrence in expectation. The one-step inequality (47) is a deterministic consequence of the update and holds pathwise with the realized gradients:

$$\Theta_{t+1} \leq \Theta_t - \eta_t \langle \hat{F}(z_t), z_t - z^* \rangle + \eta_t^2 G^2.$$

Taking $\mathbb{E}[\cdot | \mathcal{F}_t]$ and using that z_t, z^* are $\mathcal{F}_t$ -measurable,

$$\mathbb{E}[\Theta_{t+1} | \mathcal{F}_t] \leq \Theta_t - \eta_t \langle F(z_t), z_t - z^* \rangle + \eta_t^2 G^2,$$

and the deterministic decomposition from the proof of Theorem 2 (variational inequality plus Lemma 1) bounds the middle term to give $\mathbb{E}[\Theta_{t+1} | \mathcal{F}_t] \leq (1 - m\eta_t)\Theta_t + \eta_t^2 G^2$. The tower property yields the same scalar recurrence for $\mathbb{E}[\Theta_t]$, which the induction of Theorem 2 solves to $\mathbb{E}[\Theta_T] \leq M/(T + t_0)$. Each round uses one Bernoulli query per coordinate: $|Y|$ queries against the sampled pair (k_t, a_t) and $K|A|$ queries at the sampled response y_t . $\square$

A.7 Proof of Proposition 3 (Unbiased Regression)

Fix an ordered pair (y, y') and define $d_t(y, y') := r_t(y) - r_t(y')$. With $N = z_y - z_{y'}$, conditioning on (y, y') leaves randomness only in the sampled pair $(k, a) \sim s_t$ and the two Bernoulli draws, so

$$\begin{aligned} \mathbb{E}[N | y, y'] &= \mathbb{E}_{(k,a) \sim s_t}[P_k(y \succ a) - P_k(y' \succ a)] \\ &= d_t(y, y'). \end{aligned}$$

For any candidate π , adding and subtracting $\eta_t d_t(y, y')$ inside the square and using that the cross term vanishes in conditional expectation,

$$\begin{aligned} \mathbb{E}[(h_t(\pi; y, y') - \eta_t N)^2 | y, y'] \\ = (h_t(\pi; y, y') - \eta_t d_t(y, y'))^2 + \eta_t^2 \text{Var}(N | y, y'). \end{aligned}$$

The variance term does not depend on π , so the conditional loss is minimized exactly when $h_t(\pi; y, y') = \eta_t d_t(y, y')$, which the exact update π_{t+1} satisfies for every pair by (17); hence π_{t+1} minimizes the population loss. If π_t has full support, every ordered pair has positive sampling probability, so any other minimizer π must satisfy $h_t(\pi; y, y') = h_t(\pi_{t+1}; y, y')$ for all pairs; the μ and π_t terms cancel, leaving $\Delta_\pi(y, y') = \Delta_{\pi_{t+1}}(y, y')$ for all y, y' . Fixing y_0 and taking $y' = y_0$ shows $\log \pi(y) - \log \pi_{t+1}(y)$ is constant in y , so $\pi = C\pi_{t+1}$, and normalization forces $C = 1$. $\square$

B Adversary-Aggregation Estimators and the Objective-Stratified Target

The empirical loss (22) regresses the log-ratio term $h_t(\pi; y, y')$ onto a scalar that stands in for the exact per-pair reward gap

$$d_t(y, y') = r_t(y) - r_t(y') = \mathbb{E}_{(k,a) \sim s_t}[\hat{z}_{k,a}(y, y')], \quad (35)$$

where $\hat{z}_{k,a}(y, y') = \hat{P}_k(y \succ a) - \hat{P}_k(y' \succ a)$ is the plug-in relative preference of (20) and s_t is the adversary distribution over pairs. In the plug-in deployment $\hat{z}_{k,a}$ is a fixed function of the reward scores, so given a training pair (y, y') the only randomness in an estimate of d_t comes from how we sample or weight the pairs (k, a) . With $c_t(k, a) = \mathbb{E}_{y \sim \pi_t}[\hat{P}_k(y \succ a)]$ the pair cost of (12), the trainer's adversary is the *entropic best response* to the current cost, i.e., the fixed point of (14) under a frozen cost, equivalently one step of (14) at $\eta = 1/\kappa$ (cf. (31)):

$$s_t(k, a) \propto s_0(k, a) \exp\{-c_t(k, a)/\kappa\}. \quad (36)$$

The 1.5B pipeline reaches (36) approximately, by iterating the mirror step 25 times under the frozen cost; the SafeRLHF pipeline evaluates it in closed form. In both cases the adversary is recomputed at every policy stage rather than carried

as a coupled OMD iterate; Section discusses what the last-iterate theorem does and does not cover under this schedule. As in Section , write $\omega_t(k) = \sum_a s_t(k, a)$ for the mass that s_t places on objective k and $q_t(a | k) = s_t(k, a)/\omega_t(k)$ for the induced distribution over responses inside that objective, so that $s_t(k, a) = \omega_t(k) q_t(a | k)$.

Three targets. The trainer forms the regression target for a pair (y, y') in one of three ways, ordered here from most to least adversary mass retained:

$$\tau_t^{\text{full}}(y, y') = \sum_{k,a} s_t(k, a) \hat{z}_{k,a}(y, y') = d_t(y, y'), \quad (37)$$

$$\tau_t^{\text{os}}(y, y') = \sum_k \omega_t(k) \hat{z}_{k,a_k}(y, y'),$$

$$\tau_t^{\text{top}}(y, y') = \hat{z}_{k^*, a^*}(y, y'), \quad (38)$$

where $a_k \sim q_t(\cdot | k)$ is drawn independently for each objective and $(k^*, a^*) \in \arg \max_{k,a} s_t(k, a)$. The full-expectation target keeps every atom of s_t and equals d_t exactly, at a cost of one plug-in evaluation for each of the $K|\mathcal{A}|$ pairs. The objective-stratified target, deployed in the trained loss (22), draws the responses a_k independently across objectives; it keeps the exact objective marginals $\omega_t(k)$ but replaces the inner sum over responses by a single draw per objective, so it costs K evaluations and gives every objective a training signal on every update. Top-mass keeps only the single heaviest atom, or a few heaviest atoms renormalized to unit mass; it costs one evaluation but discards the remaining mass and is biased whenever that mass matters. The next two results make the middle ground precise: the stratified target is exactly unbiased for the full expectation, and it never has larger variance than a single sampled pair.

Proposition 4 (Unbiasedness of the stratified target). *Fix a pair (y, y') and the weights s_t . Over the independent draws $a_k \sim q_t(\cdot | k)$,*

$$\mathbb{E}[\tau_t^{\text{os}}(y, y')] = \tau_t^{\text{full}}(y, y') = d_t(y, y').$$

Proof. Linearity of expectation and independence of the draws give

$$\begin{aligned} \mathbb{E}[\tau_t^{\text{os}}] &= \sum_k \omega_t(k) \mathbb{E}_{a_k \sim q_t(\cdot | k)} [\hat{z}_{k,a_k}] \\ &= \sum_k \omega_t(k) \sum_a q_t(a | k) \hat{z}_{k,a}. \end{aligned}$$

Substituting $\omega_t(k) q_t(a | k) = s_t(k, a)$ collapses the double sum to $\sum_{k,a} s_t(k, a) \hat{z}_{k,a}$, which is $\tau_t^{\text{full}} = d_t$ by (37). $\square$

Proposition 5 (Variance reduction over single-pair sampling). *Let $\tau_t^{\text{iid}}(y, y') = \hat{z}_{k', a'}(y, y')$ use one pair $(k', a') \sim s_t$, the estimator behind the population loss (18) under the plug-in oracle. Set $\mu_k = \mathbb{E}_{a \sim q_t(\cdot | k)} [\hat{z}_{k,a}]$ and $v_k = \text{Var}_{a \sim q_t(\cdot | k)} (\hat{z}_{k,a})$. Both estimators are unbiased for d_t , and*

$$\begin{aligned} \text{Var}(\tau_t^{\text{iid}}) - \text{Var}(\tau_t^{\text{os}}) &= \sum_k \omega_t(k) (1 - \omega_t(k)) v_k \\ &\quad + \sum_k \omega_t(k) (\mu_k - d_t)^2 \geq 0. \end{aligned}$$

Equality holds only when every within-objective variance v_k is zero and all objective means μ_k coincide.

Proof. Unbiasedness of τ_t^{os} is Proposition 4; for τ_t^{iid} it is immediate from $\mathbb{E}[\hat{z}_{k', a'}] = \sum_{k,a} s_t(k, a) \hat{z}_{k,a} = d_t$. The K draws in τ_t^{os} are independent, so

$$\text{Var}(\tau_t^{\text{os}}) = \sum_k \omega_t(k)^2 \text{Var}(\hat{z}_{k,a_k}) = \sum_k \omega_t(k)^2 v_k.$$

For τ_t^{iid} , condition on the sampled objective k' , whose marginal is ω_t . Since $\mathbb{E}[\hat{z} | k' = k] = \mu_k$, $\text{Var}(\hat{z} | k' = k) = v_k$, and $\sum_k \omega_t(k) \mu_k = d_t$, the law of total variance gives

$$\begin{aligned} \text{Var}(\tau_t^{\text{iid}}) &= \mathbb{E}_{k'} [\text{Var}(\hat{z} | k')] + \text{Var}_{k'} (\mathbb{E}[\hat{z} | k']) \\ &= \sum_k \omega_t(k) v_k + \sum_k \omega_t(k) (\mu_k - d_t)^2. \end{aligned}$$

Subtracting this from $\text{Var}(\tau_t^{\text{os}}) = \sum_k \omega_t(k)^2 v_k$ and grouping the v_k terms yields

$$\begin{aligned} \text{Var}(\tau_t^{\text{iid}}) - \text{Var}(\tau_t^{\text{os}}) &= \sum_k (\omega_t(k) - \omega_t(k)^2) v_k \\ &\quad + \sum_k \omega_t(k) (\mu_k - d_t)^2. \end{aligned}$$

Both sums are nonnegative because $\omega_t(k) \in [0, 1]$ and $v_k \geq 0$, which proves the inequality and the equality condition. $\square$

The decomposition names two distinct gains. The between-objective term $\sum_k \omega_t(k) (\mu_k - d_t)^2$ is the variance a single draw pays for not knowing which objective it will land on; stratification removes it in full, and it is largest exactly when the objectives disagree, the regime RONPO is built for. The remaining term is the further gain from spending the query budget on one response per objective rather than on one pair overall. The stratified target thus keeps the unbiasedness of the full expectation while querying K preferences per update instead of $K|\mathcal{A}|$, so it stays affordable as the pool or the objective set grows. Top-mass sits at the opposite corner: cheapest, but biased and blind to every objective other than the current heaviest, which is the mechanism behind its instability at larger scale.

Estimator ablation at 1.5B. Table 4 isolates the aggregation choice. The two policies are stage-2 RONPO runs that share the base model, prompts, response pool, and every optimization setting, and differ only in whether the per-pair target is τ^{top} or τ^{os} . We score the held-out generations with the three local reward models under the stage-2 protocol of Appendix E.2, normalized per prompt within this three-policy set. The stratified target raises the worst objective by 0.016 over top-mass and by 0.086 over the base policy, improves the average objective, and holds the tight cross-objective spread, matching the ordering that Proposition 5 predicts.

C Standard Online-Mirror-Descent Facts

The building blocks below are classical (Cesa-Bianchi and Lugosi 2006; Bubeck 2015; Nemirovski and Yudin 1983); we include them so the paper is self-contained. We take Pinsker’s and Hölder’s inequalities as given and derive the regret-to-gap reduction, the one-step inequality (47), and the regret bound used in the section on no-regret learning.

Target	Avg	Worst	WR _B	wWR _B	Spread
Base	0.487	0.414	—	—	0.056
Top-mass τ^{top}	0.512	0.484	0.511	0.449	0.026
Stratified τ^{os}	0.527	0.500	0.522	0.468	0.030

Table 4: Estimator ablation on Qwen2.5-1.5B at stage 2, same pool and settings, differing only in the adversary-aggregation target. Avg and Worst are the mean and minimum per-prompt min-max-normalized reward across the three objectives; WR_B and wWR_B are the average and worst-objective win rate against the base response; Spread is the cross-objective standard deviation. Scores are normalized within this three-policy set and are not comparable to Table 2. The objective-stratified target gives the best average and worst-objective reward.

C.1 Entropic OMD on the Simplex

For convex losses with subgradients g_t at the iterate u_t , the entropic OMD step is

$$u_{t+1} = \arg \min_{v \in \Delta(\mathcal{Y})} \{ \eta_t \langle g_t, v \rangle + \text{KL}(v \| u_t) \}, \quad (39)$$

whose closed form is the multiplicative-weights update

$$u_{t+1}(y) = \frac{u_t(y) \exp(-\eta_t g_t(y))}{\sum_a u_t(a) \exp(-\eta_t g_t(a))}, \quad (40)$$

derived in Appendix C.4, with tuned-constant-step regret $R_T \leq G \sqrt{2T \log |\mathcal{Y}|}$ for $\|g_t\|_\infty \leq G$ (Appendix C.3).

C.2 Regret-to-Gap Reduction

At round t , let the maximizing player use π_t and the minimizing player use π'_t in the game (2). Once π'_t is fixed, the maximizing player faces the convex online loss $\ell_t^\pi(\pi) := -J(\pi, \pi'_t)$; once π_t is fixed, the minimizing player faces $\ell_t^{\pi'}(\pi') := J(\pi_t, \pi')$. Convexity holds because $J(\cdot, \pi'_t)$ is concave and $J(\pi_t, \cdot)$ is convex. The external regrets are

$$R_T^\pi = \max_{\pi \in \Delta(\mathcal{Y})} \sum_{t=1}^T J(\pi, \pi'_t) - \sum_{t=1}^T J(\pi_t, \pi'_t),$$

$$R_T^{\pi'} = \sum_{t=1}^T J(\pi_t, \pi'_t) - \min_{\pi' \in \Delta(\mathcal{Y})} \sum_{t=1}^T J(\pi_t, \pi').$$

Let $\bar{\pi}_T = \frac{1}{T} \sum_t \pi_t$ and $\bar{\pi}'_T = \frac{1}{T} \sum_t \pi'_t$, and define the gap $\text{Gap}(\bar{\pi}_T, \bar{\pi}'_T) = \max_\pi J(\pi, \bar{\pi}'_T) - \min_{\pi'} J(\bar{\pi}_T, \pi')$. For fixed π , convexity of $J(\pi, \cdot)$ and Jensen give $J(\pi, \bar{\pi}'_T) \leq \frac{1}{T} \sum_t J(\pi, \pi'_t)$; for fixed π' , concavity gives $J(\bar{\pi}_T, \pi') \geq \frac{1}{T} \sum_t J(\pi_t, \pi')$. Taking the maximum in the first, the minimum in the second, and adding and subtracting $\sum_t J(\pi_t, \pi'_t)$,

$$\text{Gap}(\bar{\pi}_T, \bar{\pi}'_T) \leq \frac{R_T^\pi + R_T^{\pi'}}{T}. \quad (41)$$

Thus if both players are no-regret, the gap of average play vanishes; this is the precise game-theoretic reason for using no-regret dynamics rather than arbitrary stabilized best responses. Sublinear regret certifies average iterates only; RONPO's last-iterate results use the additional strong monotonicity of Lemma 1.

C.3 One-Step Inequality and Regret Bound

Let $\ell_1, \dots, \ell_T$ be convex losses on $\Delta(\mathcal{Y})$, let u_{t+1} be the entropic-OMD iterate (39) from u_t with step $\eta_t > 0$, let $g_t \in \partial \ell_t(u_t)$, and fix a comparator $\tilde{u} \in \Delta(\mathcal{Y})$; write $n = |\mathcal{Y}|$.

Step 0 (Linearization). By the subgradient inequality, $\ell_t(u_t) - \ell_t(\tilde{u}) \leq \langle g_t, u_t - \tilde{u} \rangle$ (equality for linear losses, as in the RONPO blocks); summing, it suffices to bound each $\langle g_t, u_t - \tilde{u} \rangle$.

Step 1 (Closed form and log-linear relation). The subproblem (39) has the multiplicative-weights closed form (40) (App. C.4); taking logarithms,

$$\eta_t g_t = \log(u_t / u_{t+1}) - (\log Z_t) \mathbf{1} \quad (42)$$

componentwise, with all logarithms finite since interiority is preserved.

Step 2 (Three-point identity). Pair (42) with $u_{t+1} - \tilde{u}$; the constant drops because $\langle \mathbf{1}, u_{t+1} - \tilde{u} \rangle = 0$. Splitting the resulting sum into its u_{t+1} - and $\tilde{u}$ -weighted parts and using $\log(u_t / u_{t+1}) = \log(\tilde{u} / u_{t+1}) - \log(\tilde{u} / u_t)$,

$$\eta_t \langle g_t, u_{t+1} - \tilde{u} \rangle = \text{KL}(\tilde{u} \| u_t) - \text{KL}(\tilde{u} \| u_{t+1}) - \text{KL}(u_{t+1} \| u_t). \quad (43)$$

Step 3 (Splitting the per-step term). Inserting u_{t+1} as an intermediate point and substituting (43),

$$\eta_t \langle g_t, u_t - \tilde{u} \rangle = \eta_t \langle g_t, u_t - u_{t+1} \rangle + \text{KL}(\tilde{u} \| u_t) - \text{KL}(\tilde{u} \| u_{t+1}) - \text{KL}(u_{t+1} \| u_t). \quad (44)$$

Step 4 (Hölder, Pinsker, Young). Hölder gives $\eta_t \langle g_t, u_t - u_{t+1} \rangle \leq \eta_t \|g_t\|_\infty \|u_t - u_{t+1}\|_1$ and Pinsker gives $-\text{KL}(u_{t+1} \| u_t) \leq -\frac{1}{2} \|u_t - u_{t+1}\|_1^2$; with $a = \eta_t \|g_t\|_\infty$ and $b = \|u_t - u_{t+1}\|_1$, the elementary bound $ab - \frac{1}{2} b^2 \leq \frac{1}{2} a^2$ yields

$$\eta_t \langle g_t, u_t - u_{t+1} \rangle - \text{KL}(u_{t+1} \| u_t) \leq \frac{1}{2} \eta_t^2 \|g_t\|_\infty^2. \quad (45)$$

Step 5 (One-step inequality). Substituting (45) into (44) and dividing by η_t ,

$$\langle g_t, u_t - \tilde{u} \rangle \leq \frac{\text{KL}(\tilde{u} \| u_t) - \text{KL}(\tilde{u} \| u_{t+1})}{\eta_t} + \frac{\eta_t}{2} \|g_t\|_\infty^2; \quad (46)$$

multiplying back by η_t and rearranging gives the descent form

$$\text{KL}(\tilde{u} \| u_{t+1}) \leq \text{KL}(\tilde{u} \| u_t) - \eta_t \langle g_t, u_t - \tilde{u} \rangle + \frac{1}{2} \eta_t^2 \|g_t\|_\infty^2, \quad (47)$$

the per-block step invoked in Theorem 2 and Corollary 1.

Step 6 (Regret at constant step). With $\eta_t \equiv \eta$, summing (46) telescopes the KL terms; using $\text{KL}(\tilde{u} \| u_1) \leq \log n$ for uniform u_1 and $\|g_t\|_\infty \leq G$,

$$R_T \leq \frac{\log n}{\eta} + \frac{\eta}{2} G^2 T \leq G \sqrt{2T \log n}$$

at the optimal $\eta = \sqrt{2 \log n / (G^2 T)}$, so the dynamics are no-regret.

Method	$V_{\min}$	Gap	V_{avg}	Ent.	Decoy
SPPO-avg	0.060	0.269	0.535	0.005	1.000
INPO-avg	0.210	0.119	0.530	1.806	0.290
MNPO-hist-avg	<u>0.213</u>	<u>0.116</u>	0.512	2.043	0.157
HT-MNPO-mix	0.209	0.120	0.530	1.811	0.278
RONPO	0.288	0.041	<u>0.533</u>	1.386	0.340

Table 5: Final metrics in the decoy game ($V^* = 0.3289$). “Gap” is $V^* - V_{\min}$. RONPO attains the best worst-case win rate while preserving a competitive average.

C.4 Multiplicative-Weights Mirror Step

The subproblem (39) has a strictly convex objective with a full-support reference u_t , so its minimizer is interior and unique (the entropy gradient $\log v(y)$ diverges as $v(y) \downarrow 0$, so no coordinate is set to zero). Introducing a multiplier λ for $\sum_y v(y) = 1$ gives the stationarity condition $\eta_t g_t(y) + \log(v(y)/u_t(y)) + 1 + \lambda = 0$, hence $v(y) \propto u_t(y)e^{-\eta_t g_t(y)}$; normalizing recovers (40). Since u_1 is interior and each step preserves full support, every iterate is interior.

D Synthetic-Game Details and Additional Figures

Game construction. Each objective $k \in \{1, 2, 3\}$ is an 8×8 skew-symmetric matrix built in three layers: (i) a weak random tournament background assigning each ordered pair win probability 0.56 or 0.44 uniformly at random; (ii) a specialist action a_k for objective k that beats every other action with probability 0.82 under P_k ; (iii) a decoy action that beats every other action with probability 0.78 under objectives 1, 2 and 0.06 under objective 3. The reference μ and adversary prior s_0 are uniform. The robust LP benchmark $V^* = \max_{\pi} \min_{k,a} c_{\pi}(k, a)$ is computed by linear programming; the reported instance has $V^* = 0.3289$.

Compared methods and metrics. We compare five idealized tabular methods: SPPO-avg, INPO-avg, and MNPO-hist-avg on the averaged oracle $\bar{P} = \frac{1}{3} \sum_k P_k$; HT-MNPO-mix, which trains objective-specific players and deploys their arithmetic mixture; and exact RONPO over all objective-action pairs. The central metric is the robust minimum win rate $V_{\min}(\pi) = \min_{k,a} c_{\pi}(k, a)$, compared against the robust LP benchmark $V^* = \max_{\pi} V_{\min}(\pi) = 0.3289$; we also report the average reference win rate V_{avg} , final entropy, and decoy mass.

Averaging hides the failure; the pair adversary exposes it. Table 5 and Figure 2 show the mechanism. SPPO-avg collapses almost entirely onto the decoy: its average win rate is the best of all methods while its minimum falls to 0.060. This is precisely the hidden-objective failure the construction was designed to reveal, invisible to any average-based metric. Reference anchoring (INPO-avg) and historical mixing (MNPO-hist-avg) soften but do not fix the collapse, since both still optimize the fixed averaged oracle, and HT-MNPO-mix spreads mass across players without restoring a two-player saddle. RONPO raises $V_{\min}$ to 0.288, reducing the LP

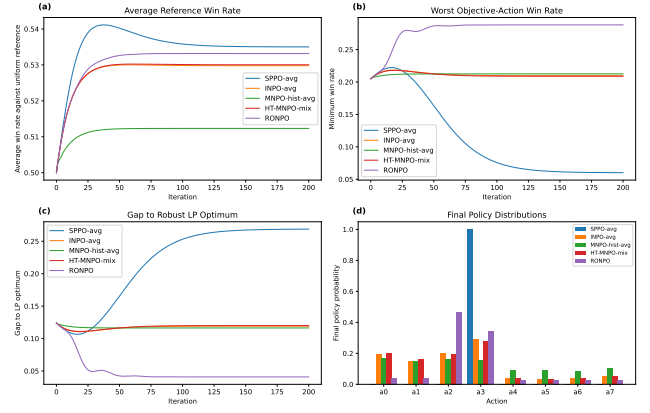


Figure 2: Dynamics over 200 iterations. Averaged-oracle SPPO attains the highest average win rate by collapsing onto the decoy action; RONPO shifts mass toward the robust LP mixture and raises the minimum objective-action win rate.

gap from 0.269 (SPPO-avg) to 0.041, at an average cost of 0.002. The final adversary concentrates on objective 3, the source that rejects the decoy, confirming that the adversary identifies exposing pairs rather than injecting noise. We further sweep the decoy severity below (RONPO’s advantage is largest exactly when the hidden failure is most severe) and the temperature κ , which interpolates between worst-case and averaging behavior as Prop. 1 predicts.

Decoy severity sweep. Figure 4 varies the hidden-objective decoy win probability. When the decoy is most dangerous, averaged-oracle methods keep assigning it high mass because it remains attractive under the first two objectives, and RONPO’s advantage is largest; as the hidden objective becomes milder, robust and average objectives align and the gap narrows.

Temperature sweep. Figure 5 traces the average/worst-case trade-off as κ varies: small κ sharpens the worst-case adversary; large κ keeps the adversary near its prior and recovers averaging behavior, the empirical counterpart of the interpolation in Prop. 1.

Random game families. The preceding panels verify algebraic statements on designed instances; Figure 3 and Table 6 separately validate the *algorithm* across random game families. For each configuration we sample 20 random tournaments (each $P_k(i, j) \sim \text{Unif}[0.05, 0.95]$, skew-symmetrized), solve the saddle to residual 10^{-13} by mirror-prox, and run the dynamics of Theorem 2. Three observations. First, the exact iterate satisfies the bound $M/(t + t_0)$ at every recorded iterate in every game, 320 of 320 games across the 16 configurations of Table 6, with a comfortable median terminal ratio (2.3×10^{-11} at the base configuration). Second, at matched exact-gradient budgets, mirror-prox (the extragradient update behind EGPO, run on the lifted game) converges linearly to numerical precision while plain OMD follows its $O(1/T)$ envelope (panel b), quantifying the rate discussion after Theorem 2. Third, single-pair Bernoulli feedback at-

Axis	Value	Ratio _T	Bound	Slope
base	–	2.3×10^{-11}	20/20	−0.98
$ \mathcal{Y} $	4	5.8×10^{-11}	20/20	−1.06
$ \mathcal{Y} $	16	1.8×10^{-12}	20/20	−1.00
$ \mathcal{Y} $	32	5.3×10^{-14}	20/20	−1.00
K	2	3.0×10^{-11}	20/20	−1.00
K	5	7.5×10^{-12}	20/20	−1.03
K	8	7.6×10^{-12}	20/20	−0.97
coverage	0.25	2.2×10^{-10}	20/20	−1.00
coverage	0.5	8.0×10^{-11}	20/20	−1.00
coverage	0.75	3.7×10^{-11}	20/20	−1.08
$\tau = \kappa$	0.01	1.1×10^{-6}	20/20	−0.87
$\tau = \kappa$	0.2	1.1×10^{-14}	20/20	−1.01
$\tau = \kappa$	0.5	1.4×10^{-14}	20/20	−1.02
noise σ	0.05	3.1×10^{-11}	20/20	−1.07
noise σ	0.1	4.5×10^{-11}	20/20	−1.01
noise σ	0.2	1.2×10^{-10}	20/20	−1.03

Table 6: One-at-a-time sensitivity from the base configuration $|\mathcal{Y}|=8$, $K=3$, coverage 1.0, $\tau=\kappa=0.05$, $\sigma=0$ (20 random games per row). Ratio_T is the median terminal ratio $\Theta_T/(M/(T+t_0))$ at $T = 5000$ for exact OMD; Bound counts games in which the Theorem 2 bound holds at every recorded iterate; Slope is the median stochastic log–log slope over the final decade (theory: −1).

tains the $\Theta(1/t)$ rate of Corollary 1, with a median log–log slope of −0.98 over the final decade and per-configuration medians within $[-1.08, -0.87]$. Panel (d) extends the value comparison beyond the designed decoy instance: on 20 uniform tournaments and 20 decoy games with randomized parameters, the lifted saddle policy leaves a smaller robust LP gap than the averaged-oracle saddle, with paired median differences of 0.035 (95% bootstrap CI $[0.027, 0.040]$) on uniform games and 0.090 ($[0.068, 0.104]$) on decoy games. The separation is largest exactly when a hidden objective conflicts with the average, matching the mechanism of Table 5; the sensitivity table shows the bound and the stochastic rate are stable across $|\mathcal{Y}|$, K , pool coverage, $\tau = \kappa$, and payoff noise.

E LLM Experiment Details

E.1 Stage-1 Evaluation

Compared models and training signals. All methods are initialized from Qwen/Qwen2.5-1.5B-Instruct. HT-MNPO and RONPO share the same stage-1 base-policy response pool; SPPO-avg and INPO-avg use the prompt-wise average of the min–max-normalized objectives.

Training configuration. Table 7 summarizes the common stage-1 configuration for HT-MNPO and RONPO; Table 8 gives method-specific settings. The SPPO-avg and INPO-avg artifacts record the checkpoint and oracle construction but not the complete optimizer configuration; we do not infer unrecorded hyperparameters. No common hyperparameter search was run across all methods: HT-MNPO and RONPO use the fixed settings above, while SPPO-avg and INPO-avg were retrained under a stability-gated sweep after their

original runs failed the generation gate, and the surviving configurations are the ones reported. We state this asymmetry, and the absence of per-method attempt counts for the original artifacts, as a protocol limitation of the 1.5B study.

Evaluation configuration and metrics. For evaluation, every model generates one response per prompt under identical vLLM decoding (seed 42, temperature 0.7, top- p 0.9, at most 2048 new tokens), and all outputs are rescored by the same three reward models in one pass (Table 10). Let $r_{i,o,m}$ be the score on prompt i from objective o for model m . Because reward scales differ, we report per-prompt min–max-normalized scores

$$\hat{r}_{i,o,m} = \frac{r_{i,o,m} - \min_{m'} r_{i,o,m'}}{\max_{m'} r_{i,o,m'} - \min_{m'} r_{i,o,m'}}, \quad (48)$$

with $\hat{r}_{i,o,m} = 1/2$ when the denominator is zero, averaged over prompts; the extrema range over the exact compared model set, so normalized values are comparable only within one table. Win rates against the base use $w = 1$ if $r_{i,o,m} > r_{i,o,m_0}$, $w = 0$ if smaller, and $w = 1/2$ on exact ties. Aggregates are computed at the objective level: Avg is the mean of the three objective-level means and Worst is their minimum, the worst objective mean of the Experiments section (this differs from the per-prompt worst used on SafeRLHF); the normalized standard deviation is computed across the three objective-level normalized scores.

E.2 Stage-2 Evaluation

Compared artifacts and setup. Stage 2 resumes training from stage-1 checkpoints, comparing the base model, the three single-oracle HT-MNPO checkpoints, and RONPO (an earlier diagnostic RONPO stage-2 run is excluded because it exhibited output-length pathology; the resumed run reported here does not). The exclusion criterion is deterministic, generation-level length pathology detected before any reward evaluation, and the excluded run together with the repaired averaged-oracle baselines of Appendix E.1 are the only reruns in this study; because retry budgets were not pre-registered uniformly across methods, we flag residual survivorship risk explicitly rather than claiming a fully symmetric selection protocol. The local reward-model and GPT-5.5 evaluations share the same generated responses, so they differ only in the evaluator; IFEval measures rule-based instruction following on a separate benchmark with deterministic decoding.

External-judge protocol. The judge receives the prompt and two candidate responses labeled A and B, in a deterministically randomized order seeded by the prompt hash, and returns a single preference (A, B, or tie) under a fixed rubric covering helpfulness, correctness, and instruction following. Each unordered pair is judged once per prompt; a tie counts as 0.5 for both sides. Mean pairwise win rates and 95% intervals use 2000 prompt-level paired bootstrap resamples.

E.3 SafeRLHF Helpfulness–Harmlessness Experiment

This appendix documents the model-scale SafeRLHF runs summarized in Table 3. Every method starts from

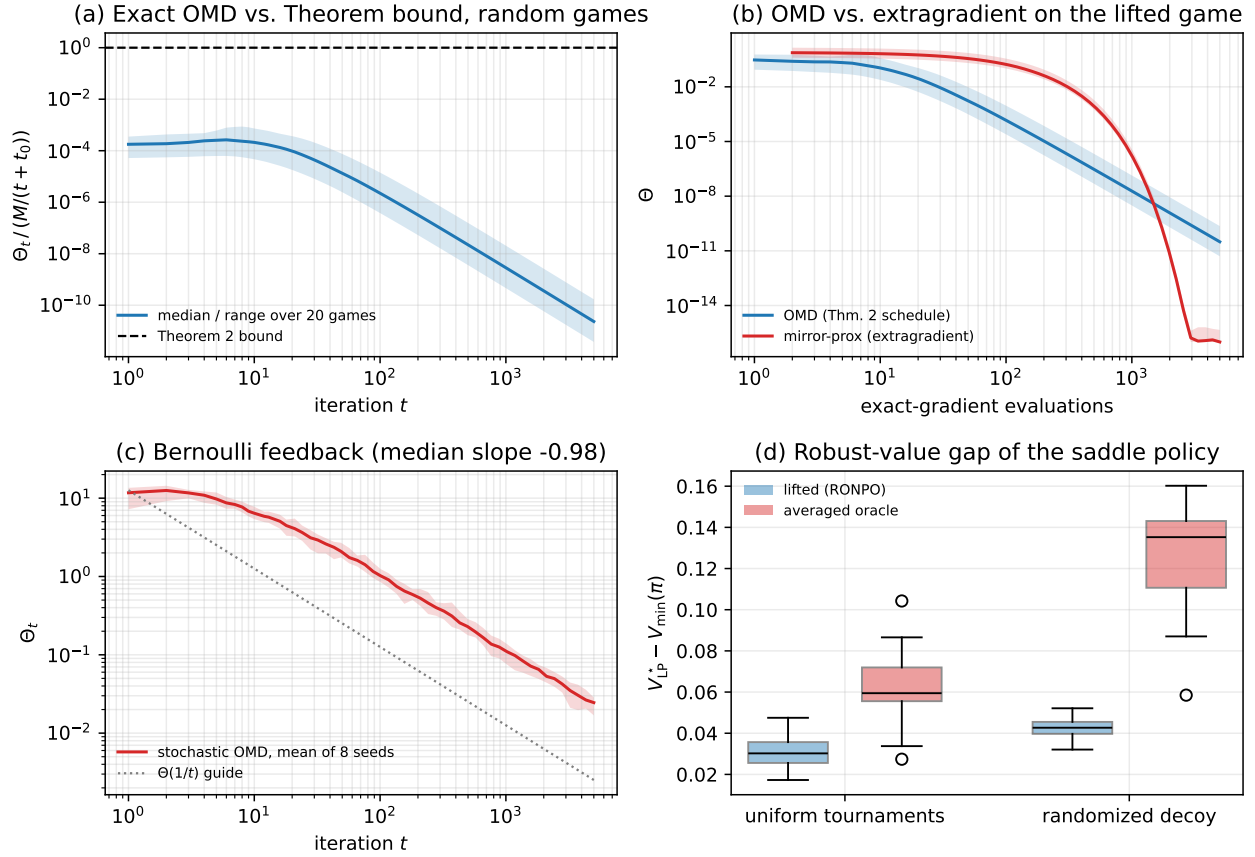


Figure 3: Validation across random game families (20 games per configuration; bands span the min–max range across games, lines are medians). (a) Exact-OMD Lyapunov value relative to the Theorem 2 bound; the bound (dashed) holds in every game. (b) Exact OMD vs. mirror-prox (extragradient) on the lifted game at matched exact-gradient budgets: mirror-prox is linearly convergent, OMD follows $O(1/T)$. (c) Single-pair Bernoulli feedback, mean of 8 seeds per game (Cor. 1). (d) Robust-value gap $V_{LP}^* - V_{\min}(\pi)$ of the saddle policy for the lifted game vs. the averaged oracle, on uniform tournaments and randomized decoy games.

Llama-3.1-8B-Instruct and trains for four iterative stages: stage 1 optimizes from the base policy, and stages 2–4 each resume from the previous stage while keeping the base policy as the reference anchor. All methods share the base policy, the training prompts, and the response pools, so the comparison isolates the training signal. Table 12 lists the configuration common to every method and stage, Table 13 the method-specific settings, and Table 14 the evaluation protocol.

E.4 Qwen3-8B Instruction-Following Scale-Up

This diagnostic tests whether the instruction-following preservation seen at 1.5B survives a larger policy, training on Qwen3-8B under three heterogeneous ArmoRM objectives (helpfulness, safety, and conciseness; Appendix E.5). Table 15 reports the four official IFEval aggregations. Because the reported configurations were selected on this same benchmark, the table is validation evidence, not test evidence, and the between-method differences are small; we therefore

draw no comparative conclusion from it and use it only as a sanity check that preference optimization at this scale does not damage instruction following. A clean protocol, selecting configurations on a separate validation split and running the official test once, is deferred to the revision. This study also uses a different base model from the SafeRLHF experiment of Appendix E.3, so the IFEval numbers here and the reward-based results there do not describe a single 8B policy.

E.5 Qwen3-8B Heterogeneous-Objective Diagnostic

As a second model-scale probe, we ran a Qwen3-8B diagnostic on a sealed held-out test with three conflicting ArmoRM objectives (helpfulness, safety, conciseness). The study is underpowered: across the ten stable policies the mean raw rewards span at most 0.0045 per objective, so per-prompt normalization amplifies statistically indistinguishable differences and the resulting ordering diagnoses evaluator power, not model robustness; pre-committed rescoring with two gen-

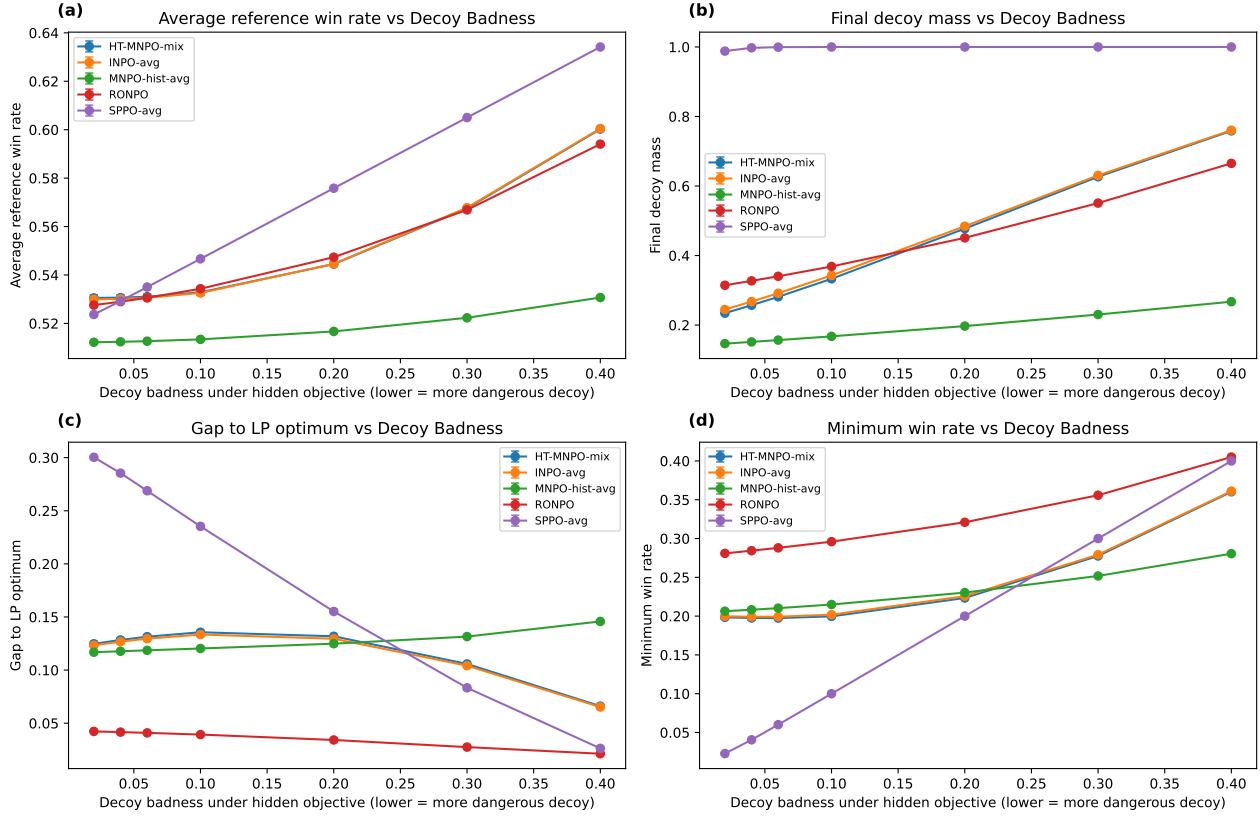


Figure 4: Sweep over the hidden-objective decoy win probability $P_3(a_{\text{decoy}} \succ a)$. Lower values make the decoy more dangerous. RONPO is most useful in the regime where the averaged oracle masks severe minority-objective failure.

eral reward models likewise separates no trained policy from base. We record this as honest negative evidence, defer the full diagnostic to the supplementary material, and will precede any future scale-up with an a priori evaluator power analysis.

E.6 Qwen3-8B Capability Preservation

We check that preference optimization at the 8B backbone does not damage general capability, using a shared seed-42 academic suite under lm-evaluation-harness 0.4.12 with thinking disabled and fixed per-task shot counts. Table 16 reports the accessible benchmarks with a macro average and its signed difference from the base policy. The macro is the mean of the six benchmarks shown. GPQA is omitted because the official dataset is access-gated, and HumanEval is excluded from the macro entirely: a think-mode chat-template failure prevents the harness from extracting code for every model, and a broken task does not belong in a capability aggregate even as a constant offset; we will rerun it once the extraction is fixed. All runs use a single seed and most per-task deltas are within run-to-run noise, so we read the table as showing no evidence of capability damage rather than as a ranking among methods.

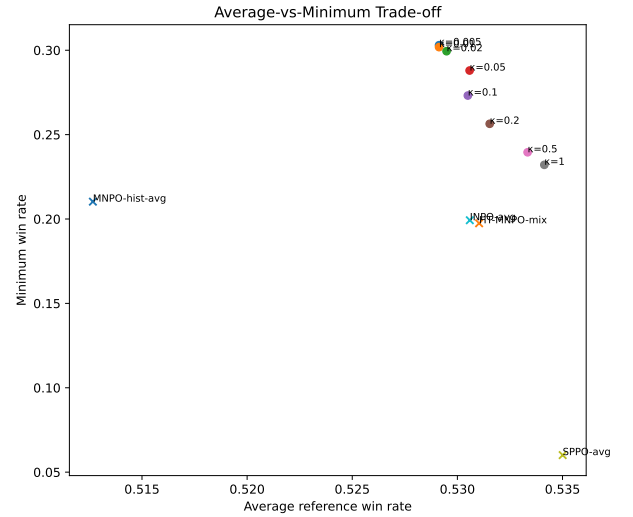


Figure 5: RONPO trade-off as the adversary temperature κ varies. Smaller κ gives a sharper worst-case adversary; larger κ moves toward fixed scalarization.

Setting	Value
Base policy	Qwen/Qwen2.5-1.5B-Instruct
Training split	UltraFeedback subset; 19,856 prompts
Held-out split	UltraFeedback subset; 647 prompts
Online generation	vLLM with seeds 13, 21, 42, 79, 100, temperature 0.8, top- p 0.95, at most 4096 new tokens
Objectives	Skywork, Athene, and ArmoRM
Shared stage-1 data	HT-MNPO and RONPO use the same base-policy response pool
Optimization	bf16 AdamW, learning rate 5×10^{-7} , cosine schedule, warmup ratio 0.1, weight decay 0
Training horizon	One epoch; seed 42
Batching	Per-device batch size 2; gradient accumulation 8
Sequence limits	Maximum sequence length 2048; maximum prompt length 1800
Memory and launch	Gradient checkpointing (<code>use_reentrant=false</code>); Accelerate with DeepSpeed ZeRO-3

Table 7: Common stage-1 training configuration for HT-MNPO and RONPO.

Setting	HT-MNPO stage 1	RONPO stage 1
Objective handling	One homogeneous oracle per run	Skywork, Athene, and ArmoRM jointly
Pair or target construction	Reward-gap target; no target normalization	Per-objective min-max normalization; adversarial pair selection; best-vs-adversary policy pair
Method scales	Ratio 0.3333; $\eta = 0.0075$; $\beta = (10, 1, 10)$ for Skywork, Athene, ArmoRM	Policy $\alpha = 1.0$; $\tau = 0.05$; preference scale $\lambda = 8.0$; $\beta = 10$
Adversary settings	Not applicable	25 adversary steps; adversary $\alpha = 1.0$; $\kappa = 0.05$; one pair per prompt

Table 8: Method-specific stage-1 training configuration.

Method	Skywork		Athene		ArmoRM		Normalized aggregate			WR aggregate	
	Norm.	WR	Norm.	WR	Norm.	WR	Avg.	Worst	Std.	Avg.	Worst
Base	0.3990	–	0.4116	–	0.4846	–	0.4317	0.3990	0.0378	–	–
HT-MNPO (Sky.)	0.4666	55.95	0.4813	58.19	0.5529	58.73	0.5002	0.4666	0.0377	57.62	55.95
HT-MNPO (Ath.)	0.4749	56.03	0.4225	49.69	0.4880	49.61	0.4618	0.4225	0.0283	51.78	49.61
HT-MNPO (Armo)	0.4615	56.34	0.4356	55.72	0.4806	50.70	0.4592	0.4356	<u>0.0184</u>	54.25	50.70
RONPO	0.6331	67.47	<u>0.6432</u>	68.55	0.6339	<u>59.81</u>	0.6367	0.6331	0.0046	65.28	<u>59.81</u>
SPPO-avg	0.5131	59.51	0.4194	52.94	0.4583	48.92	0.4636	0.4194	0.0384	53.79	48.92
INPO-avg	<u>0.5962</u>	<u>64.30</u>	0.6768	<u>67.31</u>	<u>0.6208</u>	60.43	<u>0.6313</u>	<u>0.5962</u>	0.0337	<u>64.01</u>	60.43

Table 9: Stage-1 local reward-model evaluation. *Norm.* is the per-prompt normalized reward and *WR* the win rate against the base response (%). Lower normalized standard deviation indicates more balanced objective-level performance. RONPO attains the best average and worst-objective normalized reward and the most balanced objective profile; the repaired INPO-avg is the strongest baseline and edges RONPO on the Athene objective.

Objective	Reward model
Skywork	Skywork/Skywork-Reward-V2-Llama-3.1-8B
Athene	Nexusflow/Athene-RM-8B
ArmoRM	RLHFlow/ArmoRM-Llama3-8B-v0.1

Table 10: The three heterogeneous reward models used as the stage-1 objectives.

Method	Mean pairwise WR	95% CI
RONPO	0.5927	[0.5709, 0.6136]
Base	0.5581	[0.5329, 0.5841]
HT-MNPO (Sky.)	0.5054	[0.4841, 0.5272]
HT-MNPO (Armo)	0.4263	[0.4025, 0.4493]
HT-MNPO (Ath.)	0.3643	[0.3425, 0.3855]

Table 11: Reward-model-independent GPT-5.5 all-pairwise judge results on 647 held-out prompts. Each of the five methods is compared against the other four; the averaged-oracle baselines are not part of this tournament (see Baselines). Higher mean pairwise win rate is better; the per-cell head-to-head matrix accompanies the code release.

Setting	Value
Base policy	meta-llama/Llama-3.1-8B-Instruct
Training data	PKU-SafeRLHF (Dai et al. 2024), train split restricted to the human dual-preference conflicts in which the more helpful response is the less safe one; 2,500 prompts per stage
Objectives	Helpfulness from the Beaver reward model PKU-Alignment/beaver-7b-v1.0-reward; harmlessness from the negated Beaver cost model PKU-Alignment/beaver-7b-v1.0-cost
Response pool	vLLM, four responses per prompt; stage 1 from the base policy (seeds 42–45), stages 2–4 mix parent-policy (seeds 42, 43) and base (seeds 44, 45) responses; temperature 0.7, top- p 0.9, at most 512 new tokens, bf16
Pair construction	Three pairs per prompt, selected deterministically; the preferred response has the higher mean of the two per-prompt min–max-normalized objectives; preference scale $\lambda = 8.0$
Optimization	bf16 AdamW, learning rate 5×10^{-7} , cosine schedule, warmup ratio 0.1, weight decay 0, gradient clipping 1.0
Training horizon	900 optimizer steps per stage; one independent four-stage run per training seed in {42, 43, 44}. The training seed drives data ordering and optimizer stochasticity; decode seeds are unchanged across runs, yet the parent-policy portions of the stage-2–4 pools still differ across training seeds because the parent policies differ. Evaluation decoding and bootstrap seeds stay fixed at 42
Batching	Per-device batch size 1; gradient accumulation 16
Sequence limits	Maximum sequence length 2048; maximum prompt length 1024
Regularizers	Reference-anchor weight 0.05; preference-SFT weight 0.005
Memory and launch	Gradient checkpointing (use_reentrant=false); single-GPU Accelerate

Table 12: Common training configuration for the SafeRLHF experiment.

Method	Loss	Method-specific settings
RONPO	ronpo	Objective-stratified target at adversary temperature $\kappa = 0.1$; policy $\alpha = 1.0$; $\tau = 0.05$; closed-form entropic adversary over the two objectives, one opponent sampled per objective
HT-MNPO (help./harml.)	ht_mnpo	One homogeneous oracle per run (helpfulness or harmlessness); reward-gap target; $\eta = 0.0075$
INPO-avg	inpo	Prompt-wise average of the normalized objectives; $\eta = 0.0075$; ratio 0.3333
SPPO-avg	sppo	Prompt-wise average of the normalized objectives; per-response win-rate target; $\eta = 0.0075$
SimPO	simpo	$\beta = 2.0$; $\gamma = 0.6$
IPO	ipo	$\beta = 0.05$; squared log-ratio loss
DPO	dpo	$\beta = 0.05$; length-normalized

Table 13: Method-specific training configuration for the SafeRLHF experiment. All methods share the common configuration in Table 12.

Setting	Value
Evaluation panel	1,000 PKU-SafeRLHF prompts from the official default test split, prompt-disjoint from training and from every intermediate split
Generation	vLLM, seed 42, temperature 0.7, top- p 0.9, at most 512 new tokens, bf16
Scoring	Every response rescored by both Beaver heads; helpfulness is the reward and harmlessness is the negated cost
Normalization	Per-prompt min–max over the compared policies; Avg and Worst take the mean and minimum across the two objectives
Stability gate	Fail-closed per policy: 1,000 non-empty responses, no leaked reasoning spans, mean length within $[0.33, 2.0]$ of the base policy, and at most 20 consecutive identical words
Confidence intervals	2,000 prompt-level paired bootstrap resamples, seed 42

Table 14: Evaluation configuration for the SafeRLHF experiment.

Method	Strict		Loose	
	Prompt	Inst.	Prompt	Inst.
Base	0.8336	0.8900	0.8651	0.9140
DPO	0.8281	0.8808	0.8688	0.9122
SPPO	0.8281	0.8811	0.8614	0.9036
SimPO	0.8466	<u>0.8943</u>	<u>0.8799</u>	<u>0.9187</u>
RONPO	<u>0.8410</u>	0.8962	0.8817	0.9276

Table 15: Qwen3-8B scale-up diagnostic on all IFEval prompts, reporting all four official aggregations. Rows are configurations selected on this same benchmark from fixed method sweeps, so the table is a sanity check of instruction-following preservation, not an untouched-test comparison.

Method	IFEval	MMLU	MMLU-Pro	ARC	GSM8K	AIME	Macro Δ
Base	80.96	72.03	63.16	63.74	81.65	26.67	64.70 (± 0)
RONPO (top-mass)	82.44	71.99	63.07	63.65	80.97	30.00	65.35 (+0.65)
RONPO (full-exp.)	81.52	71.97	63.00	63.57	80.89	23.33	64.05 (-0.66)
DPO	81.33	71.87	63.51	63.48	81.20	30.00	65.23 (+0.53)
IPO	82.44	71.88	63.20	63.57	81.12	33.33	65.92 (+1.22)
SimPO	80.41	71.92	63.07	63.57	81.05	23.33	63.89 (-0.81)
SPPO (avg)	81.15	71.74	63.41	63.74	80.29	30.00	65.05 (+0.35)
INPO (avg)	81.52	71.83	63.31	63.65	80.52	23.33	64.03 (-0.68)
HT-MNPO (help.)	81.15	72.08	63.39	63.65	80.52	23.33	64.02 (-0.68)
HT-MNPO (safety)	81.89	71.89	63.37	63.65	80.89	33.33	65.84 (+1.13)
HT-MNPO (concise)	81.89	71.71	63.57	63.23	80.59	23.33	64.05 (-0.65)

Table 16: **Capability preservation** on the shared seed-42 academic suite (lm-evaluation-harness 0.4.12, think disabled). Macro Δ is the mean over the six displayed benchmarks with the signed difference from base in parentheses; GPQA is access-gated, and HumanEval is excluded because a think-mode code-extraction failure invalidates it for every model. Single seed; per-benchmark deltas are within run-to-run noise (AIME-24 has 30 items, so $\pm 3.3\text{pp}$ equals one problem), so the table supports a no-evidence-of-regression reading rather than a ranking.